

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

1 Introduction

Language model pre-training has been shown to be effective for improving many natural language processing tasks (Dai and Le, 2015; Peters et al., 2018a; Radford et al., 2018; Howard and Ruder, 2018). These include sentence-level tasks such as natural language inference (Bowman et al., 2015; Williams et al., 2018) and paraphrasing (Dolan and Brockett, 2005), which aim to predict the relationships between sentences by analyzing them holistically, as well as token-level tasks such as named entity recognition and question answering, where models are required to produce fine-grained output at the token level (Tjong Kim Sang and De Meulder, 2003; Rajpurkar et al., 2016).

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers of the Transformer (Vaswani et al., 2017). Such restrictions are sub-optimal for sentence-level tasks, and could be very harmful when applying fine-tuning based approaches to token-level tasks such as question answering, where it is crucial to incorporate context from both directions.

In this paper, we improve the fine-tuning based approaches by proposing BERT: **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. BERT alleviates the previously mentioned unidirectionality constraint by using a “masked language model” (MLM) pre-training objective, inspired by the Cloze task (Taylor, 1953). The masked language model randomly masks some of the tokens from the input, and the objective is to predict the original vocabulary id of the masked

BERT：用于语言理解的深度双向 Transformer 预训练

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摘要

我们提出了一种新的语言表示模型，名为 **BERT**，其全称是双向 **编码器表示** 来自 Transformers。与近期的语言表示模型（Peters 等人，2018a；Radford 等人，2018）不同，BERT 旨在通过在所有层中同时以左右上下文为条件，从未标注文本中预训练深度双向表示。因此，只需添加一个输出层，就能对预训练的 BERT 模型进行微调，从而为问答和语言推理等各种任务构建达到最先进水平的模型，而无需大幅修改特定任务的架构。

BERT 在概念上简单明了，在实证中表现强劲。它在 11 项自然语言处理任务上取得了新的最先进成果，包括将 GLUE 得分提升至 80.5%（绝对提高 7.7 个百分点），将 MultiNLI 准确率提升至 86.7%（绝对提高 4.6 个百分点），将 SQuAD v1.1 问答测试集 F1 提升至 93.2（绝对提高 1.5 个百分点），并将 SQuAD v2.0 测试集 F1 提升至 83.1（绝对提高 5.1 个百分点）。

1 引言

语言模型预训练已被证明能够有效提升许多自然语言处理任务的性能（Dai and Le, 2015；Peters et al., 2018a；Radford et al., 2018；Howard and Ruder, 2018）。这些任务包括句子级任务，例如自然语言推理（Bowman et al., 2015；Williams et al., 2018）和释义识别（Dolan and Brockett, 2005），其目标是通过整体分析句子来预测句子之间的关系；此外还包括词元级任务，例如命名实体识别和问答。在这些任务中，模型需要在词元级别生成细粒度输出（Tjong Kim Sang and De Meulder, 2003；Rajpurkar et al., 2016）。

将预训练语言表示应用于下游任务，现有两种策略：特征提取和微调。特征提取方法（例如 ELMo）（Peters et al., 2018a）采用特定于任务的架构，将预训练表示作为附加特征纳入其中。微调方法（例如生成式预训练 Transformer（OpenAI GPT））（Radford et al., 2018）引入的特定于任务的参数极少，并通过对所有预训练参数进行简单微调，在下游任务上训练模型。这两种方法在预训练期间使用相同的目标函数，通过单向语言模型学习通用语言表示。

我们认为，当前技术限制了预训练表示的能力，尤其是对于微调方法而言。其主要局限在于，标准语言模型是单向的，这限制了预训练阶段可使用的架构选择。例如，在 OpenAI GPT 中，作者采用从左到右的架构，在 Transformer 的自注意力层中，每个词元只能关注之前的词元（Vaswani 等人，2017）。这类限制并非最优，尤其不适用于句子级任务；当将基于微调的方法用于问答等词元级任务时，这种限制可能造成严重影响，因为此类任务至关重要的一点是结合来自两个方向的上下文。

在本文中，我们通过提出 BERT：**双向编码器表示（基于Transformer）**。BERT 通过采用受完形填空任务启发的“掩码语言模型”（MLM）预训练目标，缓解了前文所述的单向性约束（Taylor, 1953）。掩码语言模型会随机遮蔽输入中的部分词元，其目标是预测被遮蔽词元原本对应的词表索引

word based only on its context. Unlike left-to-right language model pre-training, the MLM objective enables the representation to fuse the left and the right context, which allows us to pre-train a deep bidirectional Transformer. In addition to the masked language model, we also use a “next sentence prediction” task that jointly pre-trains text-pair representations. The contributions of our paper are as follows:

- We demonstrate the importance of bidirectional pre-training for language representations. Unlike Radford et al. (2018), which uses unidirectional language models for pre-training, BERT uses masked language models to enable pre-trained deep bidirectional representations. This is also in contrast to Peters et al. (2018a), which uses a shallow concatenation of independently trained left-to-right and right-to-left LMs.
- We show that pre-trained representations reduce the need for many heavily-engineered task-specific architectures. BERT is the first fine-tuning based representation model that achieves state-of-the-art performance on a large suite of sentence-level and token-level tasks, outperforming many task-specific architectures.
- BERT advances the state of the art for eleven NLP tasks. The code and pre-trained models are available at <https://github.com/google-research/bert>.

2 Related Work

There is a long history of pre-training general language representations, and we briefly review the most widely-used approaches in this section.

2.1 Unsupervised Feature-based Approaches

Learning widely applicable representations of words has been an active area of research for decades, including non-neural (Brown et al., 1992; Ando and Zhang, 2005; Blitzer et al., 2006) and neural (Mikolov et al., 2013; Pennington et al., 2014) methods. Pre-trained word embeddings are an integral part of modern NLP systems, offering significant improvements over embeddings learned from scratch (Turian et al., 2010). To pre-train word embedding vectors, left-to-right language modeling objectives have been used (Mnih and Hinton, 2009), as well as objectives to discriminate correct from incorrect words in left and right context (Mikolov et al., 2013).

These approaches have been generalized to coarser granularities, such as sentence embeddings (Kiros et al., 2015; Logeswaran and Lee, 2018) or paragraph embeddings (Le and Mikolov, 2014). To train sentence representations, prior work has used objectives to rank candidate next sentences (Jernite et al., 2017; Logeswaran and Lee, 2018), left-to-right generation of next sentence words given a representation of the previous sentence (Kiros et al., 2015), or denoising auto-encoder derived objectives (Hill et al., 2016).

ELMo and its predecessor (Peters et al., 2017, 2018a) generalize traditional word embedding research along a different dimension. They extract *context-sensitive* features from a left-to-right and a right-to-left language model. The contextual representation of each token is the concatenation of the left-to-right and right-to-left representations. When integrating contextual word embeddings with existing task-specific architectures, ELMo advances the state of the art for several major NLP benchmarks (Peters et al., 2018a) including question answering (Rajpurkar et al., 2016), sentiment analysis (Socher et al., 2013), and named entity recognition (Tjong Kim Sang and De Meulder, 2003). Melamud et al. (2016) proposed learning contextual representations through a task to predict a single word from both left and right context using LSTMs. Similar to ELMo, their model is feature-based and not deeply bidirectional. Fedus et al. (2018) shows that the cloze task can be used to improve the robustness of text generation models.

2.2 Unsupervised Fine-tuning Approaches

As with the feature-based approaches, the first works in this direction only pre-trained word embedding parameters from unlabeled text (Collobert and Weston, 2008).

More recently, sentence or document encoders which produce contextual token representations have been pre-trained from unlabeled text and fine-tuned for a supervised downstream task (Dai and Le, 2015; Howard and Ruder, 2018; Radford et al., 2018). The advantage of these approaches is that few parameters need to be learned from scratch. At least partly due to this advantage, OpenAI GPT (Radford et al., 2018) achieved previously state-of-the-art results on many sentence-level tasks from the GLUE benchmark (Wang et al., 2018a). Left-to-right language model-

, and only rely on its context. Unlike left-to-right language model pre-training, the MLM objective enables the representation to fuse the left and the right context, which allows us to pre-train a deep bidirectional Transformer. In addition to the masked language model, we also use a “next sentence prediction” task that jointly pre-trains text-pair representations. The contributions of our paper are as follows:

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- BERT advances the state of the art for eleven NLP tasks. The code and pre-trained models are available at <https://github.com/google-research/bert>.

2 Related Work

通用语言表示的预训练已有悠久历史，本节将简要回顾应用最广泛的方法。

2.1 无监督的基于特征的方法

数十年来，学习广泛适用的词语表征一直是一个活跃的研究领域，涵盖非神经网络方法（Brown et al., 1992; Ando and Zhang, 2005; Blitzer et al., 2006）以及神经网络方法（Mikolov et al., 2013; Pennington et al., 2014）。预训练词嵌入是现代自然语言处理系统的重要组成部分，与从头开始学习的嵌入相比，能够带来显著提升（Turian et al., 2010）。为预训练词嵌入向量，研究者采用了从左到右的语言建模目标（Mnih and Hinton, 2009），也采用了用于区分左右文中正确词语与错误词语的目标（Mikolov et al., 2013）。

这些方法已被推广到更粗粒度的层面，例如句子嵌入（Kiros et al., 2015; Logeswaran and Lee, 2018）或段落嵌入（Le and Mikolov, 2014）。为训练句子表征，先前的研究采用过对候选后续句子进行排序的目标（Jernite et al., 2017; Logeswaran and Lee, 2018）、给定前一句子的表征从左到右生成后一句子中的词语（Kiros et al., 2015），或基于去噪自编码器构造的目标（Hillet et al., 2016）。

ELMo 及其前身（Peters et al., 2017, 2018a）沿着另一个维度拓展了传统的词嵌入研究。它们从从左到右和从右到左的语言模型中提取上下文相关特征。每个词元的上下文表示由从左到右和从右到左的表示拼接而成。将上下文词嵌入与现有的特定任务架构相结合时，ELMo 在多个重要 NLP 基准测试中取得了最先进的结果（Peters et al., 2018a），包括问答（Rajpurkar et al., 2016）、情感分析（Socher et al., 2013）和命名实体识别（Tjong Kim Sang and De Meulder, 2003）。Melamud et al. (2016) 提出使用 LSTM，通过一项同时利用左右文预测单个词的任务来学习上下文表示。与 ELMo 类似，他们的模型基于特征，而非深度双向模型。Fedus et al. (2018) 表明，完形填空任务可用于提升文本生成模型的鲁棒性。

2.2 无监督微调方法

与基于特征的方法一样，最初沿着这一方向开展的工作只对来自未标注文本的词嵌入参数进行预训练（Collobert and Weston, 2008）。

近年来，能够生成上下文词元表示的句子或文档编码器，已通过无标注文本进行预训练，并针对有监督的下游任务进行微调（Dai and Le, 2015; Howard and Ruder, 2018; Radford et al., 2018）。这些方法的优势在于，只需从头学习少量参数。至少部分得益于这一优势，OpenAI GPT（Radford et al., 2018）在 GLUE 基准（Wang et al., 2018a）中的许多句子级任务上取得了此前最先进的结果。左右语言建模

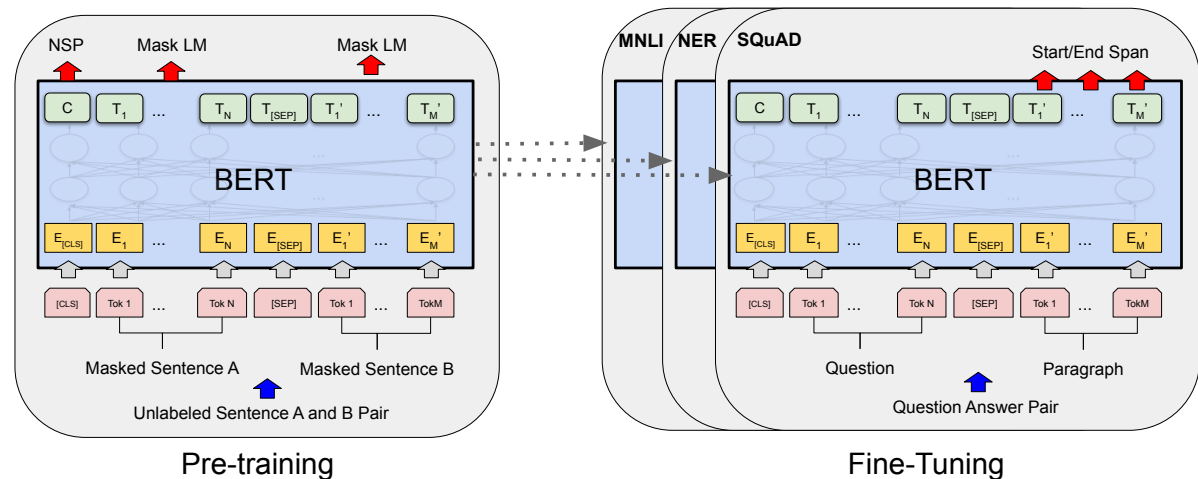


Figure 1: Overall pre-training and fine-tuning procedures for BERT. Apart from output layers, the same architectures are used in both pre-training and fine-tuning. The same pre-trained model parameters are used to initialize models for different down-stream tasks. During fine-tuning, all parameters are fine-tuned. [CLS] is a special symbol added in front of every input example, and [SEP] is a special separator token (e.g. separating questions/answers).

ing and auto-encoder objectives have been used for pre-training such models (Howard and Ruder, 2018; Radford et al., 2018; Dai and Le, 2015).

2.3 Transfer Learning from Supervised Data

There has also been work showing effective transfer from supervised tasks with large datasets, such as natural language inference (Conneau et al., 2017) and machine translation (McCann et al., 2017). Computer vision research has also demonstrated the importance of transfer learning from large pre-trained models, where an effective recipe is to fine-tune models pre-trained with ImageNet (Deng et al., 2009; Yosinski et al., 2014).

3 BERT

We introduce BERT and its detailed implementation in this section. There are two steps in our framework: *pre-training* and *fine-tuning*. During pre-training, the model is trained on unlabeled data over different pre-training tasks. For fine-tuning, the BERT model is first initialized with the pre-trained parameters, and all of the parameters are fine-tuned using labeled data from the downstream tasks. Each downstream task has separate fine-tuned models, even though they are initialized with the same pre-trained parameters. The question-answering example in Figure 1 will serve as a running example for this section.

A distinctive feature of BERT is its unified architecture across different tasks. There is mini-

mal difference between the pre-trained architecture and the final downstream architecture.

Model Architecture BERT’s model architecture is a multi-layer bidirectional Transformer encoder based on the original implementation described in Vaswani et al. (2017) and released in the `tensor2tensor` library.¹ Because the use of Transformers has become common and our implementation is almost identical to the original, we will omit an exhaustive background description of the model architecture and refer readers to Vaswani et al. (2017) as well as excellent guides such as “The Annotated Transformer.”²

In this work, we denote the number of layers (i.e., Transformer blocks) as L , the hidden size as H , and the number of self-attention heads as A .³ We primarily report results on two model sizes: **BERT_{BASE}** ($L=12$, $H=768$, $A=12$, Total Parameters=110M) and **BERT_{LARGE}** ($L=24$, $H=1024$, $A=16$, Total Parameters=340M).

BERT_{BASE} was chosen to have the same model size as OpenAI GPT for comparison purposes. Critically, however, the BERT Transformer uses bidirectional self-attention, while the GPT Transformer uses constrained self-attention where every token can only attend to context to its left.⁴

¹<https://github.com/tensorflow/tensor2tensor>

²<http://nlp.seas.harvard.edu/2018/04/03/attention.html>

³In all cases we set the feed-forward/filter size to be $4H$, i.e., 3072 for the $H = 768$ and 4096 for the $H = 1024$.

⁴We note that in the literature the bidirectional Trans-

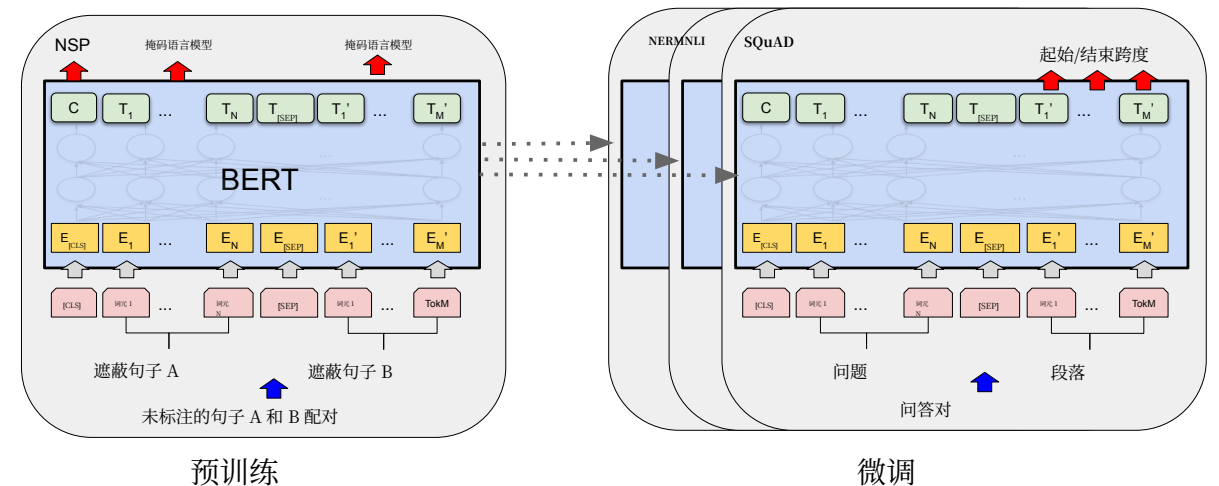


图 1: BERT 的整体预训练和微调流程。除输出层外，预训练和微调使用相同的架构。使用相同的预训练模型参数来初始化针对不同下游任务的模型。在微调期间，所有参数都会进行微调。[CLS] 是添加在每个输入示例前面的特殊符号，[SEP] 是特殊分隔标记（例如，用于分隔问题和答案）。

以及自编码器目标已被用于预训练此类模型

(Howard and Ruder, 2018; Radford et al., 2018; Dai and Le, 2015)。

2.3 基于监督数据的迁移学习

也有研究表明，从拥有大型数据集的监督任务中进行迁移是有效的，例如自然语言推理 (Conneau et al., 2017) 和机器翻译 (McCann et al., 2017)。计算机视觉研究也证明了迁移学习的重要性：对于大型预训练模型，一种有效的方法是微调在 ImageNet 上预训练的模型 (Denget al., 2009; Yosinski et al., 2014)。

3 BERT

本节将介绍 BERT 及其详细实现。我们的框架包含两个步骤：预训练和微调。在预训练阶段，模型会针对不同的预训练任务，使用无标注数据进行训练。在微调阶段，BERT 模型首先使用预训练参数进行初始化，然后使用下游任务的标注数据对所有参数进行微调。每个下游任务都有各自经过微调的模型，尽管这些模型都使用相同的预训练参数进行初始化。图 1 中的问答示例将作为本节贯穿始终的示例。

BERT 的一个显著特点是，它在不同任务中采用统一的架构。几乎没有

预训练架构与最终下游架构之间的差异。

模型架构 BERT 的模型架构是一种多层双向 Transformer 编码器，基于 Vaswani 等人 (2017) 在 `tensor2tensor` 库中发布的原始实现。¹ 由于 Transformer 的使用已变得普遍，而且我们的实现与原始实现几乎完全相同，因此我们将省略对模型架构的详尽背景介绍，并请读者参阅 Vaswani 等人 (2017) 以及《The Annotated Transformer》等优秀指南。²

在本文中，我们将层数（即 Transformer 模块数）记为 L ，将隐藏层大小记为 H ，将自注意力头数记为 A 。³ 我们主要报告两种模型规模的结果：BERT_{BASE} ($L=12$, $H=768$, $A=12$, 总参数量=110M) 和 BERT_{LARGE} ($L=24$, $H=1024$, $A=16$, 总参数量=340M)。

BERT_{BASE} 被设计为与 OpenAI GPT 具有相同的模型规模，以便进行比较。然而，关键区别在于，BERT Transformer 使用双向自注意力，而 GPT Transformer 使用受限自注意力，每个词元只能关注其左侧的上下文。⁴

¹<https://github.com/tensorflow/tensor2tensor>

²<http://nlp.seas.harvard.edu/2018/04/03/attention.html>³所有情况下，我们都将前馈层/滤波器的大小设为 $4H$ ，即 $H = 768$ 为 3072，

$H = 1024$ 为 4096。⁴我们注意到，在文献中，双向 Trans-

Input/Output Representations To make BERT handle a variety of down-stream tasks, our input representation is able to unambiguously represent both a single sentence and a pair of sentences (e.g., $\langle \text{Question, Answer} \rangle$) in one token sequence. Throughout this work, a “sentence” can be an arbitrary span of contiguous text, rather than an actual linguistic sentence. A “sequence” refers to the input token sequence to BERT, which may be a single sentence or two sentences packed together.

We use WordPiece embeddings (Wu et al., 2016) with a 30,000 token vocabulary. The first token of every sequence is always a special classification token ($[\text{CLS}]$). The final hidden state corresponding to this token is used as the aggregate sequence representation for classification tasks. Sentence pairs are packed together into a single sequence. We differentiate the sentences in two ways. First, we separate them with a special token ($[\text{SEP}]$). Second, we add a learned embedding to every token indicating whether it belongs to sentence A or sentence B. As shown in Figure 1, we denote input embedding as E , the final hidden vector of the special $[\text{CLS}]$ token as $C \in \mathbb{R}^H$, and the final hidden vector for the i^{th} input token as $T_i \in \mathbb{R}^H$.

For a given token, its input representation is constructed by summing the corresponding token, segment, and position embeddings. A visualization of this construction can be seen in Figure 2.

3.1 Pre-training BERT

Unlike Peters et al. (2018a) and Radford et al. (2018), we do not use traditional left-to-right or right-to-left language models to pre-train BERT. Instead, we pre-train BERT using two unsupervised tasks, described in this section. This step is presented in the left part of Figure 1.

Task #1: Masked LM Intuitively, it is reasonable to believe that a deep bidirectional model is strictly more powerful than either a left-to-right model or the shallow concatenation of a left-to-right and a right-to-left model. Unfortunately, standard conditional language models can only be trained left-to-right *or* right-to-left, since bidirectional conditioning would allow each word to indirectly “see itself”, and the model could trivially predict the target word in a multi-layered context.

former is often referred to as a “Transformer encoder” while the left-context-only version is referred to as a “Transformer decoder” since it can be used for text generation.

In order to train a deep bidirectional representation, we simply mask some percentage of the input tokens at random, and then predict those masked tokens. We refer to this procedure as a “masked LM” (MLM), although it is often referred to as a *Cloze* task in the literature (Taylor, 1953). In this case, the final hidden vectors corresponding to the mask tokens are fed into an output softmax over the vocabulary, as in a standard LM. In all of our experiments, we mask 15% of all WordPiece tokens in each sequence at random. In contrast to denoising auto-encoders (Vincent et al., 2008), we only predict the masked words rather than reconstructing the entire input.

Although this allows us to obtain a bidirectional pre-trained model, a downside is that we are creating a mismatch between pre-training and fine-tuning, since the $[\text{MASK}]$ token does not appear during fine-tuning. To mitigate this, we do not always replace “masked” words with the actual $[\text{MASK}]$ token. The training data generator chooses 15% of the token positions at random for prediction. If the i -th token is chosen, we replace the i -th token with (1) the $[\text{MASK}]$ token 80% of the time (2) a random token 10% of the time (3) the unchanged i -th token 10% of the time. Then, T_i will be used to predict the original token with cross entropy loss. We compare variations of this procedure in Appendix C.2.

Task #2: Next Sentence Prediction (NSP)

Many important downstream tasks such as Question Answering (QA) and Natural Language Inference (NLI) are based on understanding the *relationship* between two sentences, which is not directly captured by language modeling. In order to train a model that understands sentence relationships, we pre-train for a binarized *next sentence prediction* task that can be trivially generated from any monolingual corpus. Specifically, when choosing the sentences A and B for each pre-training example, 50% of the time B is the actual next sentence that follows A (labeled as IsNext), and 50% of the time it is a random sentence from the corpus (labeled as NotNext). As we show in Figure 1, C is used for next sentence prediction (NSP).⁵ Despite its simplicity, we demonstrate in Section 5.1 that pre-training towards this task is very beneficial to both QA and NLI.⁶

⁵The final model achieves 97%-98% accuracy on NSP.

⁶The vector C is not a meaningful sentence representation without fine-tuning, since it was trained with NSP.

输入/输出表示为了使 BERT 能够处理各种下游任务，我们的输入表示能够在一个词元序列中明确表示单个句子和句子对（例如， $\langle \text{问题, 答案} \rangle$ ）。在本文中，“句子”可以是任意一段连续文本，而不一定是真正的语言学意义上的句子。“序列”指输入给 BERT 的词元序列，可以是单个句子，也可以是打包在一起的两个句子。

我们使用 WordPiece 嵌入 (Wu 等人, 2016)，词汇表包含 30,000 个词元。每个序列的第一个词元始终是一个特殊的分类词元 ($[\text{CLS}]$)。与该词元对应的最终隐状态被用作分类任务的整体序列表示。句子对被打包到一个序列中。我们通过两种方式区分这些句子。首先，我们用一个特殊词元 ($[\text{SEP}]$)。其次，我们为每个词元添加一个学习得到的嵌入，以指示它属于句子A 还是句子B。如图 1所示，我们将输入嵌入表示为 E ，特殊 $[\text{CLS}]$ 词元对应的最终隐向量表示为 $C \in \mathbb{R}^H$ ，而 i^{th} 输入词元对应的最终隐向量表示为 $T_i \in \mathbb{R}^H$ 。

对于给定的词元，其输入表示由对应的词元、分段和位置嵌入相加构成。图2展示了这一构造过程。

3.1 BERT 的预训练

与Peters 等人(2018a) 和Radford 等人(2018)，我们没有使用传统的从左到右或从右到左语言模型来预训练 BERT。相反，我们使用本节中介绍的两种无监督任务来预训练 BERT。这一步展示在图1。

任务#1: 掩码语言模型 直观而言，有理由认为，深度双向模型的能力严格强于单向（从左到右）模型，或将从左到右与从右到左模型浅层拼接而成的模型。遗憾的是，标准条件语言模型只能按从左到右或从右到左的方向训练，因为双向条件化会使每个词都能间接地“看到自身”，模型便可以在多层上下文中轻易预测目标词。

前者通常被称为“Transformer 编码器”，而仅使用左侧上下文的版本则被称为“Transformer 解码器”，因为它可用于文本生成。

为了训练深度双向表示，我们只需随机屏蔽一定比例的输入词元，然后预测这些被屏蔽的词元。我们将此过程称为“掩码语言模型” (MLM)，尽管在文献中它通常被称为完形填空任务 (Taylor, 1953)。在这种情况下，与掩码词元对应的最终隐藏向量会像标准语言模型一样输入词汇表上的输出 softmax。在所有实验中，我们都会在每个序列中随机屏蔽 15% 的所有 WordPiece 词元。与去噪自动编码器 (Vincent 等人, 2008) 不同，我们只预测被屏蔽的词，而不是重建整个输入。

尽管这使我们能够获得双向预训练模型，但其缺点是，我们在预训练和微调之间造成了不匹配，因为 $[\text{MASK}]$ 标记在微调期间不会出现。为缓解这一问题，我们并不总是用实际的 $[\text{MASK}]$ 标记替换“被掩盖”的词。训练数据生成器会随机选择 15% 的 token 位置进行预测。如果选择了第 i 个 token，我们会将第 i 个 token 替换为 (1) $[\text{MASK}]$ 标记 (占 80%) (2) 随机 token (占 10%) 或 (3) 保持原样的第 i 个 token (占 10%)。然后， T_i 将用于通过交叉熵损失预测原始 token。我们在附录 C.2。

任务#2: 下一句预测 (NSP) 许多重要的下游任务，例如问答 (QA) 和自然语言推理 (NLI)，都基于对关系两个句子之间关系的理解，而语言建模无法直接捕捉这种关系。为了训练能够理解句子关系的模型，我们针对一种二元分类的下一句预测任务进行预训练，这项任务可以从任何单语语料库中轻松生成。具体来说，在为每个预训练示例选择句子A 和 B 时，有 50% 的情况下，B 是紧接在A 之后的实际下一句（标记为 IsNext ），另外 50% 的情况下，它是从语料库中随机抽取的句子（标记为 NotNext ）。如图1， C 用于下一句预测 (NSP)。⁵ 尽管这项任务很简单，我们在第5.1 节中表明，针对这项任务进行预训练对问答和自然语言推理都非常有益。⁶

⁵最终模型在 NSP 上的准确率达到 97%-98%。⁶向量 C 在未经微调时并不能表示有意义的句子，因为它通过 NSP 训练的。

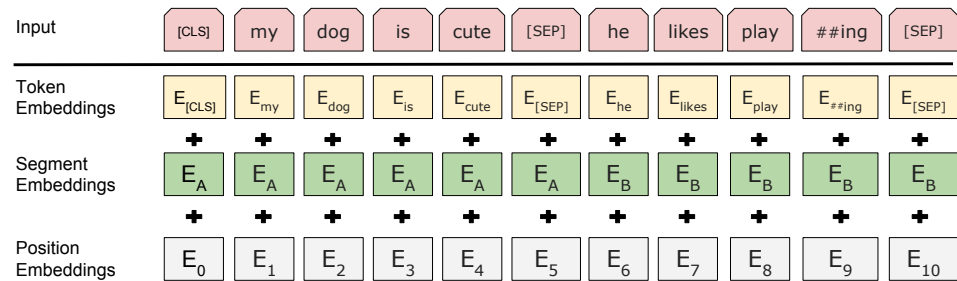


Figure 2: BERT input representation. The input embeddings are the sum of the token embeddings, the segmentation embeddings and the position embeddings.

The NSP task is closely related to representation-learning objectives used in [Jernite et al. \(2017\)](#) and [Logeswaran and Lee \(2018\)](#). However, in prior work, only sentence embeddings are transferred to down-stream tasks, where BERT transfers all parameters to initialize end-task model parameters.

Pre-training data The pre-training procedure largely follows the existing literature on language model pre-training. For the pre-training corpus we use the BooksCorpus (800M words) ([Zhu et al., 2015](#)) and English Wikipedia (2,500M words). For Wikipedia we extract only the text passages and ignore lists, tables, and headers. It is critical to use a document-level corpus rather than a shuffled sentence-level corpus such as the Billion Word Benchmark ([Chelba et al., 2013](#)) in order to extract long contiguous sequences.

3.2 Fine-tuning BERT

Fine-tuning is straightforward since the self-attention mechanism in the Transformer allows BERT to model many downstream tasks—whether they involve single text or text pairs—by swapping out the appropriate inputs and outputs. For applications involving text pairs, a common pattern is to independently encode text pairs before applying bidirectional cross attention, such as [Parikh et al. \(2016\)](#); [Seo et al. \(2017\)](#). BERT instead uses the self-attention mechanism to unify these two stages, as encoding a concatenated text pair with self-attention effectively includes *bidirectional* cross attention between two sentences.

For each task, we simply plug in the task-specific inputs and outputs into BERT and fine-tune all the parameters end-to-end. At the input, sentence A and sentence B from pre-training are analogous to (1) sentence pairs in paraphrasing, (2) hypothesis-premise pairs in entailment, (3) question-passage pairs in question answering, and

(4) a degenerate text- $\emptyset$ pair in text classification or sequence tagging. At the output, the token representations are fed into an output layer for token-level tasks, such as sequence tagging or question answering, and the [CLS] representation is fed into an output layer for classification, such as entailment or sentiment analysis.

Compared to pre-training, fine-tuning is relatively inexpensive. All of the results in the paper can be replicated in at most 1 hour on a single Cloud TPU, or a few hours on a GPU, starting from the exact same pre-trained model.⁷ We describe the task-specific details in the corresponding subsections of Section 4. More details can be found in Appendix A.5.

4 Experiments

In this section, we present BERT fine-tuning results on 11 NLP tasks.

4.1 GLUE

The General Language Understanding Evaluation (GLUE) benchmark ([Wang et al., 2018a](#)) is a collection of diverse natural language understanding tasks. Detailed descriptions of GLUE datasets are included in Appendix B.1.

To fine-tune on GLUE, we represent the input sequence (for single sentence or sentence pairs) as described in Section 3, and use the final hidden vector $C \in \mathbb{R}^H$ corresponding to the first input token ([CLS]) as the aggregate representation. The only new parameters introduced during fine-tuning are classification layer weights $W \in \mathbb{R}^{K \times H}$, where K is the number of labels. We compute a standard classification loss with C and W , i.e., $\log(\text{softmax}(CW^T))$.

⁷For example, the BERT SQuAD model can be trained in around 30 minutes on a single Cloud TPU to achieve a Dev F1 score of 91.0%.

⁸See (10) in <https://gluebenchmark.com/faq>.

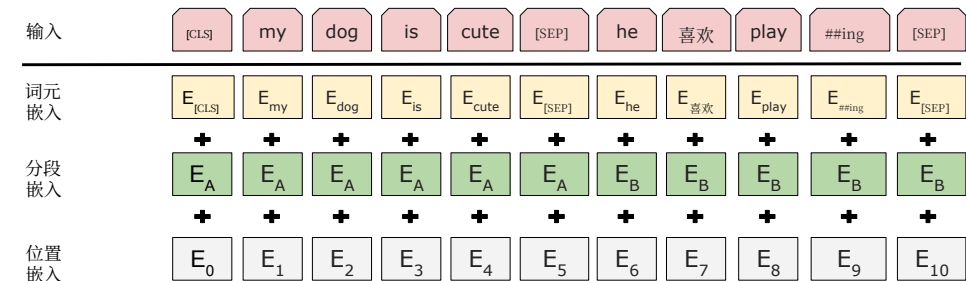


图 2: BERT 输入表示。输入嵌入是词元嵌入、分段嵌入和位置嵌入之和。

NSP 任务与表征学习目标密切相关，这类目标曾用于 [Jernite 等人 \(2017\)](#) 以及 [Logeswaran 和 Lee \(2018\)](#)。不过，在此前的工作中，只有句子嵌入被迁移到下游任务中，而 BERT 则将所有参数迁移过来，用于初始化最终任务模型的参数。

预训练数据 预训练过程大体遵循了语言模型预训练领域的现有研究。预训练语料采用 BooksCorpus (8 亿词) ([Zhu 等人, 2015](#)) 和英文维基百科 (25 亿词)。对于维基百科，我们只提取文本段落，忽略列表、表格和标题。为了提取较长的连续序列，使用文档级语料至关重要，而不能使用句子级打乱语料，例如 Billion Word Benchmark ([Chelba 等人, 2013](#))。

3.2 BERT 微调

微调过程很简单，因为 Transformer 中的自注意力机制使 BERT 能够对许多下游任务进行建模——无论这些任务涉及单个文本还是文本对——只需替换相应的输入和输出即可。对于涉及文本对的应用，一种常见做法是在应用双向交叉注意力之前，分别对文本对进行编码，例如 [Parikh 等人 \(2016\)](#); [Seo 等人 \(2017\)](#)。BERT 则使用自注意力机制将这两个阶段统一起来，因为对拼接后的文本对进行自注意力编码，实际上包含了两个句子之间的双向交叉注意力。

对于每项任务，我们只需将该任务特定的输入和输出接入 BERT，并对所有参数进行端到端微调。在输入端，预训练中的句子 A 和句子 B 分别对应于：(1) 释义任务中的句子对；(2) 蕴含任务中的假设-前提对；(3) 问答任务中的问题-篇章对；以及

(4) 文本分类或序列标注中的退化文本- $\emptyset$ 对。在输出端，对于序列标注或问答等词元级任务，将词元表示输入输出层；对于蕴含或情感分析等分类任务，则将 [CLS] 表示输入输出层。

与预训练相比，微调的成本相对较低。本文中的所有结果均可从完全相同的预训练模型出发，在单个 Cloud TPU 上用不超过 1 小时复现，或在 GPU 上用几小时复现。⁷ 具体任务的细节将在第 4 节相应的小节中进行说明。更多详情请参见附录 A.5。

4 实验

本节介绍 BERT 在 11 项 NLP 任务上的微调结果。

4.1 GLUE

通用语言理解评估 (GLUE) 基准 ([Wang 等人, 2018a](#)) 包含多种不同的自然语言理解任务。GLUE 数据集的详细介绍见附录 B.1。

为在 GLUE 上进行微调，我们按照第 3 节所述表示输入序列 (单句或句子对)，并使用最终的隐藏向量 $C \in \mathbb{R}^H$ ，该向量对应于第一个输入词元 ([CLS]) 作为聚合表示。微调过程中引入的唯一新参数是分类层权重 $W \in \mathbb{R}^{K \times H}$ ，其中 K 是标签数量。我们使用 C 和 W 计算标准分类损失，即 $\log(\text{softmax}(CW^T))$ 。

⁷例如，BERT SQuAD 模型可在单个 Cloud TPU 上训练约 30 分钟，Dev F1 分数可达到 91.0%。⁸请参见 (10)，详情请访问 <https://gluebenchmark.com/faq>。

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Table 1: GLUE Test results, scored by the evaluation server (<https://gluebenchmark.com/leaderboard>). The number below each task denotes the number of training examples. The ‘‘Average’’ column is slightly different than the official GLUE score, since we exclude the problematic WNLI set.⁸ BERT and OpenAI GPT are single-model, single task. F1 scores are reported for QQP and MRPC, Spearman correlations are reported for STS-B, and accuracy scores are reported for the other tasks. We exclude entries that use BERT as one of their components.

We use a batch size of 32 and fine-tune for 3 epochs over the data for all GLUE tasks. For each task, we selected the best fine-tuning learning rate (among 5e-5, 4e-5, 3e-5, and 2e-5) on the Dev set. Additionally, for BERT_{LARGE} we found that fine-tuning was sometimes unstable on small datasets, so we ran several random restarts and selected the best model on the Dev set. With random restarts, we use the same pre-trained checkpoint but perform different fine-tuning data shuffling and classifier layer initialization.⁹

Results are presented in Table 1. Both BERT_{BASE} and BERT_{LARGE} outperform all systems on all tasks by a substantial margin, obtaining 4.5% and 7.0% respective average accuracy improvement over the prior state of the art. Note that BERT_{BASE} and OpenAI GPT are nearly identical in terms of model architecture apart from the attention masking. For the largest and most widely reported GLUE task, MNLI, BERT obtains a 4.6% absolute accuracy improvement. On the official GLUE leaderboard¹⁰, BERT_{LARGE} obtains a score of 80.5, compared to OpenAI GPT, which obtains 72.8 as of the date of writing.

We find that BERT_{LARGE} significantly outperforms BERT_{BASE} across all tasks, especially those with very little training data. The effect of model size is explored more thoroughly in Section 5.2.

4.2 SQuAD v1.1

The Stanford Question Answering Dataset (SQuAD v1.1) is a collection of 100k crowd-sourced question/answer pairs (Rajpurkar et al., 2016). Given a question and a passage from

⁹The GLUE data set distribution does not include the Test labels, and we only made a single GLUE evaluation server submission for each of BERT_{BASE} and BERT_{LARGE}.

¹⁰<https://gluebenchmark.com/leaderboard>

Wikipedia containing the answer, the task is to predict the answer text span in the passage.

As shown in Figure 1, in the question answering task, we represent the input question and passage as a single packed sequence, with the question using the A embedding and the passage using the B embedding. We only introduce a start vector $S \in \mathbb{R}^H$ and an end vector $E \in \mathbb{R}^H$ during fine-tuning. The probability of word i being the start of the answer span is computed as a dot product between T_i and S followed by a softmax over all of the words in the paragraph: $P_i = \frac{e^{S \cdot T_i}}{\sum_j e^{S \cdot T_j}}$. The analogous formula is used for the end of the answer span. The score of a candidate span from position i to position j is defined as $S \cdot T_i + E \cdot T_j$, and the maximum scoring span where $j \geq i$ is used as a prediction. The training objective is the sum of the log-likelihoods of the correct start and end positions. We fine-tune for 3 epochs with a learning rate of 5e-5 and a batch size of 32.

Table 2 shows top leaderboard entries as well as results from top published systems (Seo et al., 2017; Clark and Gardner, 2018; Peters et al., 2018a; Hu et al., 2018). The top results from the SQuAD leaderboard do not have up-to-date public system descriptions available,¹¹ and are allowed to use any public data when training their systems. We therefore use modest data augmentation in our system by first fine-tuning on TriviaQA (Joshi et al., 2017) before fine-tuning on SQuAD.

Our best performing system outperforms the top leaderboard system by +1.5 F1 in ensembling and +1.3 F1 as a single system. In fact, our single BERT model outperforms the top ensemble system in terms of F1 score. Without TriviaQA fine-

¹¹QANet is described in Yu et al. (2018), but the system has improved substantially after publication.

系统	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	平均值 -
OpenAI 之前的最先进水平	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+注意力	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

表 1: GLUE 测试结果，由评估服务器评分 (<https://gluebenchmark.com/leaderboard>)。每项任务下方的数字表示训练样本数量。“平均值”一栏与官方 GLUE 分数略有不同，因为我们排除了有问题任务的 WNLI 任务。⁸ BERT 和 OpenAI GPT 均为单模型、单任务。QQP 和 MRPC 报告 F1 分数，STS-B 报告 Spearman 相关系数，其他任务则报告准确率。我们排除了将 BERT 作为其组件之一的条目。

对于所有 GLUE 任务，我们使用 32 的批量大小，并在数据上微调 3 个 epoch。对于每项任务，我们在开发集上选取最佳微调学习率（从 5e-5、4e-5、3e-5 和 2e-5 中选择）。此外，对于 BERT_{LARGE}，我们发现，在小型数据集上进行微调有时不稳定，因此会使用多个随机重启，并在开发集上选取最佳模型。随机重启时，我们使用相同的预训练检查点，但采用不同的微调数据打乱方式和分类器层初始化方式。⁹

结果见表1。两种BERT_{BASE} 和 BERT_{LARGE} 在所有任务上的表现都大幅优于所有系统，平均准确率分别比此前最先进水平提高了 4.5% 和 7.0%。值得注意的是，BERT_{BASE} 与 OpenAI GPT 除注意力掩码外，在模型架构方面几乎完全相同。对于规模最大且报告最广泛的 GLUE 任务 MNLI，BERT 的绝对准确率提高了 4.6 个百分点。在 GLUE 官方排行榜上¹⁰，BERT_{LARGE} 截至撰写本文时，得分为 80.5，而 OpenAI GPT 得分为 72.8。

我们发现，BERT_{LARGE} 明显优于 BERT_{BASE} 在所有任务上的表现，尤其是在训练数据极少的任务上。第 5.2 节将更深入地探讨模型规模的影响。

4.2 SQuAD v1.1

斯坦福问答数据集 (SQuAD v1.1) 是一个由 10 万个人工众包的问题/答案对组成的集合 (Rajpurkar et al., 2016)。给定一个来自

⁹GLUE 数据集的分布不包含测试集标签，并且我们仅针对 BERT_{BASE} 和 BERT_{LARGE} 各向 GLUE 评测服务器提交了一次结果。¹⁰<https://gluebenchmark.com/leaderboard>

包含答案的维基百科文章，任务是预测该段落中的答案文本片段。

如图1所示，在问答任务中，我们将输入的问题和段落表示为一个打包后的序列，其中问题使用 A 嵌入，段落使用 B 嵌入。我们仅在微调期间引入一个起始向量 $S \in \mathbb{R}^H$ 和一个结束向量 $E \in \mathbb{R}^H$ 。词语 i 作为答案片段起始词的概率，通过 T_i 与 S 的点积计算，然后对段落中的所有词语进行 softmax 归一化：

$$P_i = \frac{e^{S \cdot T_i}}{\sum_j e^{S \cdot T_j}}$$

答案片段末尾也使用类似的公式。位置 i 到位置 j 的候选片段得分定义为 $S \cdot T_i + E \cdot T_j$ ，并将满足 $j \geq i$ 的最高分片段用作预测。训练目标是正确起始位置和结束位置的对数似然之和。我们以 5e-5 的学习率和 32 的批量大小微调 3 个 epoch。

表 2 展示了排行榜上的顶尖条目以及顶尖已发表系统的结果 (Seo et al., 2017; Clark and Gardner, 2018; Peters et al., 2018a; Hu et al., 2018)。SQuAD 排行榜上的顶尖结果没有可获取的最新公开系统描述，¹¹ 而且允许在训练系统时使用任何公开数据。因此，我们的系统采用适度的数据增强：先在 TriviaQA (Joshi et al., 2017) 上进行微调，然后再在 SQuAD 上微调。

我们的最佳系统在集成后比排行榜上的顶尖系统高出 +1.5 个 F1 分数，作为单一系统时则高出 +1.3 个 F1 分数。事实上，就 F1 分数而言，我们的单个 BERT 模型优于排行榜上表现最佳的集成系统。如果不使用 TriviaQA 微调

¹¹QANet 的相关介绍见 Yu 等人 (2018)，但该系统在发表后已得到显著改进。

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	-	-	82.3	91.2
#1 Ensemble - nlnet	-	-	86.0	91.7
#2 Ensemble - QANet	-	-	84.5	90.5
Published				
BiDAF+ELMo (Single)	-	85.6	-	85.8
R.M. Reader (Ensemble)	81.2	87.9	82.3	88.5
Ours				
BERT _{BASE} (Single)	80.8	88.5	-	-
BERT _{LARGE} (Single)	84.1	90.9	-	-
BERT _{LARGE} (Ensemble)	85.8	91.8	-	-
BERT _{LARGE} (Sgl.+TriviaQA)	84.2	91.1	85.1	91.8
BERT _{LARGE} (Ens.+TriviaQA)	86.2	92.2	87.4	93.2

Table 2: SQuAD 1.1 results. The BERT ensemble is 7x systems which use different pre-training checkpoints and fine-tuning seeds.

System	Dev		Test	
	EM	F1	EM	F1
Top Leaderboard Systems (Dec 10th, 2018)				
Human	86.3	89.0	86.9	89.5
#1 Single - MIR-MRC (F-Net)	-	-	74.8	78.0
#2 Single - nlnet	-	-	74.2	77.1
Published				
unet (Ensemble)	-	-	71.4	74.9
SLQA+ (Single)	-		71.4	74.4
Ours				
BERT _{LARGE} (Single)	78.7	81.9	80.0	83.1

Table 3: SQuAD 2.0 results. We exclude entries that use BERT as one of their components.

tuning data, we only lose 0.1-0.4 F1, still outperforming all existing systems by a wide margin.¹²

4.3 SQuAD v2.0

The SQuAD 2.0 task extends the SQuAD 1.1 problem definition by allowing for the possibility that no short answer exists in the provided paragraph, making the problem more realistic.

We use a simple approach to extend the SQuAD v1.1 BERT model for this task. We treat questions that do not have an answer as having an answer span with start and end at the [CLS] token. The probability space for the start and end answer span positions is extended to include the position of the [CLS] token. For prediction, we compare the score of the no-answer span: $s_{\text{null}} = S \cdot C + E \cdot C$ to the score of the best non-null span

¹²The TriviaQA data we used consists of paragraphs from TriviaQA-Wiki formed of the first 400 tokens in documents, that contain at least one of the provided possible answers.

System	Dev	Test
ESIM+GloVe	51.9	52.7
ESIM+ELMo	59.1	59.2
OpenAI GPT	-	78.0
BERT _{BASE}	81.6	-
BERT _{LARGE}	86.6	86.3
Human (expert) [†]	-	85.0
Human (5 annotations) [†]	-	88.0

Table 4: SWAG Dev and Test accuracies. [†]Human performance is measured with 100 samples, as reported in the SWAG paper.

$s_{i,j} = \max_{j \geq i} S \cdot T_i + E \cdot T_j$. We predict a non-null answer when $s_{i,j} > s_{\text{null}} + \tau$, where the threshold τ is selected on the dev set to maximize F1. We did not use TriviaQA data for this model. We fine-tuned for 2 epochs with a learning rate of 5e-5 and a batch size of 48.

The results compared to prior leaderboard entries and top published work (Sun et al., 2018; Wang et al., 2018b) are shown in Table 3, excluding systems that use BERT as one of their components. We observe a +5.1 F1 improvement over the previous best system.

4.4 SWAG

The Situations With Adversarial Generations (SWAG) dataset contains 113k sentence-pair completion examples that evaluate grounded common-sense inference (Zellers et al., 2018). Given a sentence, the task is to choose the most plausible continuation among four choices.

When fine-tuning on the SWAG dataset, we construct four input sequences, each containing the concatenation of the given sentence (sentence A) and a possible continuation (sentence B). The only task-specific parameters introduced is a vector whose dot product with the [CLS] token representation C denotes a score for each choice which is normalized with a softmax layer.

We fine-tune the model for 3 epochs with a learning rate of 2e-5 and a batch size of 16. Results are presented in Table 4. BERT_{LARGE} outperforms the authors’ baseline ESIM+ELMo system by +27.1% and OpenAI GPT by 8.3%.

5 Ablation Studies

In this section, we perform ablation experiments over a number of facets of BERT in order to better understand their relative importance. Additional

系统	Dev		Test	
	EM	F1	EM	F1
排行榜顶尖系统（2018年12月10日）				
人类	-	-	82.3	91.2
#1 集成模型 - nlnet	-	-	86.0	91.7
#2 集成 - QANet	-	-	84.5	90.5
已发表				
BiDAF+ELMo（单模型）	-	85.6	-	85.8
R.M. Reader（集成）	81.2	87.9	82.3	88.5
Ours				
BERT _{BASE} （单模型）	80.8	88.5	-	-
BERT _{LARGE} （单模型）	84.1	90.9	-	-
BERT _{LARGE} （集成）	85.8	91.8	-	-
BERT _{LARGE} （单模型+TriviaQA）	84.2	91.1	85.1	91.8
BERT _{LARGE} （集成+TriviaQA）	86.2	92.2	87.4	93.2

表 2: SQuAD 1.1 的结果。BERT 集成模型由 7 个系统组成，这些系统使用不同的预训练检查点和微调随机种子。

系统	Dev		Test	
	EM	F1	EM	F1
排行榜顶尖系统（2018 年 12 月 10 日）				
人类	86.3	89.0	86.9	89.5
#1 单模型 - MIR-MRC (F-Net)	-	-	74.8	78.0
#2 单模型 - nlnet	-	-	74.2	77.1
已发表				
unet（集成）	-	-	71.4	74.9
SLQA+（单模型）	-		71.4	74.4
Ours				
BERT _{LARGE} （单模型）	78.7	81.9	80.0	83.1

表 3: SQuAD 2.0 结果。我们排除了将 BERT 作为其组成部分之一的条目。

数据，我们的 F1 分数只会下降 0.1–0.4，仍以显著优势领先所有现有系统。¹²

4.3 SQuAD v2.0

SQuAD 2.0 任务扩展了 SQuAD 1.1 的问题定义，允许所提供的段落中不存在简短答案的情况，从而使问题更加贴近现实。

我们采用一种简单的方法，将 SQuAD v1.1 的 BERT 模型扩展到此任务。我们将没有答案的问题视为答案跨度的起点和终点均位于 [CLS] 标记处。起始和结束答案跨度位置的概率空间扩展为包含 [CLS] 标记的位置。对于预测，我们比较无答案跨度的分数：

$$s_{\text{null}} = S \cdot C + E \cdot C \text{ 与最佳非空跨度的分数}$$

¹²我们使用的 TriviaQA 数据由 TriviaQA-Wiki 中的段落组成，这些段落取自文档开头的 400 个词元，且至少包含一个提供的可能答案。

系统	开发集	测试集
ESIM+GloVe	51.9	52.7
ESIM+ELMo	59.1	59.2
OpenAI GPT	-	78.0
BERT _{BASE}	81.6	-
BERT _{LARGE}	86.6	86.3
人类（专家） [†]	-	85.0
人类（5 次标注） [†]	-	88.0

表 4: SWAG 验证集和测试集准确率。[†]人类表现根据 100 个样本进行测量，数据引自 SWAG 论文。

$\hat{s}_{i,j} = \text{最大值}_{j \geq i} S \cdot T_i + E \cdot T_j$ 。当 $\hat{s}_{i,j} > s_{\text{null}} + \tau$ 时，我们预测非空答案，其中阈值 τ 在开发集上选取，以最大化 F1。此模型未使用 TriviaQA 数据。我们进行了 2 个轮次的微调，学习率为 5e-5，批量大小为 48。

与排行榜此前的条目及已发表的顶尖成果（Sun et al., 2018; Wang et al., 2018b）见表3，其中不包括使用 BERT 作为组件之一的系统。我们观察到 F1 分数提高了 +5.1 分，相比之前的最佳系统。

4.4 SWAG

对抗生成情境（SWAG）数据集包含 11.3 万个句子对补全示例，用于评估基于现实情境的常识推理（Zellers 等人, 2018）。给定一个句子，任务是从四个选项中选择最合理的续写。

在 SWAG 数据集上进行微调时，我们构造四个输入序列，每个序列都由给定句子（句子 A）与一个可能的续写（句子 B）拼接而成。唯一引入的特定于任务的参数是一个向量，它与 [CLS] 标记表示的点积 C 表示每个选项的分数，并通过 softmax 层进行归一化。

我们以 2e-5 的学习率和 16 的批量大小对模型进行 3 个 epoch 的微调。结果见表 4。BERT_{LARGE} 的表现优于作者的基线 ESIM+ELMo 系统 +27.1%，并优于 OpenAI GPT 8.3%。

5 消融研究

本节中，我们针对 BERT 的若干方面进行了消融实验，以便更好地了解它们各自的重要性。更多

Tasks	Dev Set				
	MNLI-m (Acc)	QNLI (Acc)	MRPC (Acc)	SST-2 (Acc)	SQuAD (F1)
BERT _{BASE}	84.4	88.4	86.7	92.7	88.5
No NSP	83.9	84.9	86.5	92.6	87.9
LTR & No NSP	82.1	84.3	77.5	92.1	77.8
+ BiLSTM	82.1	84.1	75.7	91.6	84.9

Table 5: Ablation over the pre-training tasks using the BERT_{BASE} architecture. “No NSP” is trained without the next sentence prediction task. “LTR & No NSP” is trained as a left-to-right LM without the next sentence prediction, like OpenAI GPT. “+ BiLSTM” adds a randomly initialized BiLSTM on top of the “LTR + No NSP” model during fine-tuning.

ablation studies can be found in Appendix C.

5.1 Effect of Pre-training Tasks

We demonstrate the importance of the deep bidirectionality of BERT by evaluating two pre-training objectives using exactly the same pre-training data, fine-tuning scheme, and hyperparameters as BERT_{BASE}:

No NSP: A bidirectional model which is trained using the “masked LM” (MLM) but without the “next sentence prediction” (NSP) task.

LTR & No NSP: A left-context-only model which is trained using a standard Left-to-Right (LTR) LM, rather than an MLM. The left-only constraint was also applied at fine-tuning, because removing it introduced a pre-train/fine-tune mismatch that degraded downstream performance. Additionally, this model was pre-trained without the NSP task. This is directly comparable to OpenAI GPT, but using our larger training dataset, our input representation, and our fine-tuning scheme.

We first examine the impact brought by the NSP task. In Table 5, we show that removing NSP hurts performance significantly on QNLI, MNLI, and SQuAD 1.1. Next, we evaluate the impact of training bidirectional representations by comparing “No NSP” to “LTR & No NSP”. The LTR model performs worse than the MLM model on all tasks, with large drops on MRPC and SQuAD.

For SQuAD it is intuitively clear that a LTR model will perform poorly at token predictions, since the token-level hidden states have no right-side context. In order to make a good faith attempt at strengthening the LTR system, we added a randomly initialized BiLSTM on top. This does significantly improve results on SQuAD, but the

results are still far worse than those of the pre-trained bidirectional models. The BiLSTM hurts performance on the GLUE tasks.

We recognize that it would also be possible to train separate LTR and RTL models and represent each token as the concatenation of the two models, as ELMo does. However: (a) this is twice as expensive as a single bidirectional model; (b) this is non-intuitive for tasks like QA, since the RTL model would not be able to condition the answer on the question; (c) this it is strictly less powerful than a deep bidirectional model, since it can use both left and right context at every layer.

5.2 Effect of Model Size

In this section, we explore the effect of model size on fine-tuning task accuracy. We trained a number of BERT models with a differing number of layers, hidden units, and attention heads, while otherwise using the same hyperparameters and training procedure as described previously.

Results on selected GLUE tasks are shown in Table 6. In this table, we report the average Dev Set accuracy from 5 random restarts of fine-tuning. We can see that larger models lead to a strict accuracy improvement across all four datasets, even for MRPC which only has 3,600 labeled training examples, and is substantially different from the pre-training tasks. It is also perhaps surprising that we are able to achieve such significant improvements on top of models which are already quite large relative to the existing literature. For example, the largest Transformer explored in Vaswani et al. (2017) is (L=6, H=1024, A=16) with 100M parameters for the encoder, and the largest Transformer we have found in the literature is (L=64, H=512, A=2) with 235M parameters (Al-Rfou et al., 2018). By contrast, BERT_{BASE} contains 110M parameters and BERT_{LARGE} contains 340M parameters.

It has long been known that increasing the model size will lead to continual improvements on large-scale tasks such as machine translation and language modeling, which is demonstrated by the LM perplexity of held-out training data shown in Table 6. However, we believe that this is the first work to demonstrate convincingly that scaling to extreme model sizes also leads to large improvements on very small scale tasks, provided that the model has been sufficiently pre-trained. Peters et al. (2018b) presented

任务	开发集				
	MNLI-m (准确率)	QNLI (准确率)	MRPC (准确率)	SST-2 (准确率)	SQuAD (F1)
BERT _{BASE}	84.4	88.4	86.7	92.7	88.5
无 NSP	83.9	84.9	86.5	92.6	87.9
LTR 和无 NSP	82.1	84.3	77.5	92.1	77.8
+ BiLSTM	82.1	84.1	75.7	91.6	84.9

表 5: 使用 BERT_{BASE} 架构对预训练任务进行的消融研究。“无 NSP”表示训练时不使用下一句预测任务。“LTR 和无 NSP”表示训练时不使用下一句预测任务,而是采用从左到右的语言模型,与 OpenAI GPT 类似。“+BiLSTM”表示在微调期间,在“LTR + 无 NSP”模型之上添加一个随机初始化的 BiLSTM。

消融研究可见附录C。

5.1 预训练任务的影响

我们通过评估两个预训练目标,来证明 BERT_{BASE} 深度双向性的重要性;评估时使用的预训练数据、微调方案和超参数均与 BERT 完全相同:

无 NSP: 使用“掩码语言模型”(MLM)进行训练,但不包含“下一句预测”(NSP)任务的双向模型。

LTR 且无 NSP: 仅利用左侧上下文的模型,使用标准的从左到右(LTR)语言模型进行训练,而非 MLM。微调时也采用仅利用左侧上下文的约束,因为移除该约束会造成预训练与微调不匹配,从而降低下游任务的性能。此外,该模型在预训练时不包含 NSP 任务。它与 OpenAI GPT 直接具有可比性,但使用了更大的训练数据集、我们的输入表示和微调方案。

我们首先考察 NSP 任务带来的影响。在表 5 中,我们展示了移除 NSP 会显著降低 QNLI、MNLI 和 SQuAD 1.1 上的性能。接下来,我们通过比较“无 NSP”和“LTR 且无 NSP”,评估训练双向表示的影响。LTR 模型在所有任务上的表现都不如 MLM 模型,在 MRPC 和 SQuAD 上的性能下降尤为明显。

对于 SQuAD,显而易见,LTR 模型在预测词元时表现会很差,因为词元级隐藏状态没有右侧上下文。为了尽可能公平地尝试增强 LTR 系统,我们在其顶层添加了一个随机初始化的 BiLSTM。这确实显著提升了 SQuAD 上的结果,但

结果仍远不及预训练的双向模型。BiLSTM 会降低模型在 GLUE 任务上的表现。

我们承认,也可以分别训练从左到右(LTR)和从右到左(RTL)的模型,并像 ELMo 那样将每个词元表示为两个模型输出的拼接。不过:(a)这种做法的成本是单个双向模型的两倍;(b)对于问答等任务,这并不直观,因为 RTL 模型无法根据问题来生成答案;(c)这种方法的能力严格弱于深度双向模型,因为后者在每一层都能利用左右两侧的上文。

5.2 模型规模的影响

本节探讨模型规模对微调任务准确率的影响。我们训练了多个 BERT 模型,它们的层数、隐藏单元数和注意力头数各不相同,但其他超参数和训练流程均与前文所述相同。

部分 GLUE 任务的结果见表 6。此表报告了 5 次随机重启微调所得的开发集准确率均值。可以看出,较大的模型在全部四个数据集上都能严格提升准确率;即便是只有 3,600 个带标签训练样本、且与预训练任务存在显著差异的 MRPC 也不例外。或许令人惊讶的是,相较于现有文献中的模型,我们已经使用相当大的模型,却仍能在此基础上取得如此显著的提升。例如,Vaswani 等人(2017)所考察的最大型 Transformer 为(L=6, H=1024, A=16)其编码器有 1 亿个参数;而我们在文献中发现的最大型 Transformer 则为(L=64, H=512, A=2)有 2.35 亿个参数(Al-Rfou 等人, 2018)。相比之下,BERT_{BASE}包含 1.1 亿个参数,而 BERT_{LARGE} 包含 3.4 亿个参数。

众所周知,增大模型规模能够持续提升机器翻译和语言建模等大规模任务的表现,表中所示的留出训练数据上的语言模型困惑度也证明了这一点 6。然而,我们认为,这是首项有力证明:只要模型经过了充分的预训练,将模型规模扩展到极大,也能显著提升非常小规模任务表现的研究。Peters 等人(2018b)报告了

mixed results on the downstream task impact of increasing the pre-trained bi-LM size from two to four layers and Melamud et al. (2016) mentioned in passing that increasing hidden dimension size from 200 to 600 helped, but increasing further to 1,000 did not bring further improvements. Both of these prior works used a feature-based approach — we hypothesize that when the model is fine-tuned directly on the downstream tasks and uses only a very small number of randomly initialized additional parameters, the task-specific models can benefit from the larger, more expressive pre-trained representations even when downstream task data is very small.

5.3 Feature-based Approach with BERT

All of the BERT results presented so far have used the fine-tuning approach, where a simple classification layer is added to the pre-trained model, and all parameters are jointly fine-tuned on a downstream task. However, the feature-based approach, where fixed features are extracted from the pre-trained model, has certain advantages. First, not all tasks can be easily represented by a Transformer encoder architecture, and therefore require a task-specific model architecture to be added. Second, there are major computational benefits to pre-compute an expensive representation of the training data once and then run many experiments with cheaper models on top of this representation.

In this section, we compare the two approaches by applying BERT to the CoNLL-2003 Named Entity Recognition (NER) task (Tjong Kim Sang and De Meulder, 2003). In the input to BERT, we use a case-preserving WordPiece model, and we include the maximal document context provided by the data. Following standard practice, we formulate this as a tagging task but do not use a CRF

Hyperparams				Dev Set Accuracy		
#L	#H	#A	LM (ppl)	MNLI-m	MRPC	SST-2
3	768	12	5.84	77.9	79.8	88.4
6	768	3	5.24	80.6	82.2	90.7
6	768	12	4.68	81.9	84.8	91.3
12	768	12	3.99	84.4	86.7	92.9
12	1024	16	3.54	85.7	86.9	93.3
24	1024	16	3.23	86.6	87.8	93.7

Table 6: Ablation over BERT model size. #L = the number of layers; #H = hidden size; #A = number of attention heads. “LM (ppl)” is the masked LM perplexity of held-out training data.

System	Dev F1	Test F1
ELMo (Peters et al., 2018a)	95.7	92.2
CVT (Clark et al., 2018)	-	92.6
CSE (Akbik et al., 2018)	-	93.1
Fine-tuning approach		
BERT _{LARGE}	96.6	92.8
BERT _{BASE}	96.4	92.4
Feature-based approach (BERT _{BASE})		
Embeddings	91.0	-
Second-to-Last Hidden	95.6	-
Last Hidden	94.9	-
Weighted Sum Last Four Hidden	95.9	-
Concat Last Four Hidden	96.1	-
Weighted Sum All 12 Layers	95.5	-

Table 7: CoNLL-2003 Named Entity Recognition results. Hyperparameters were selected using the Dev set. The reported Dev and Test scores are averaged over 5 random restarts using those hyperparameters.

layer in the output. We use the representation of the first sub-token as the input to the token-level classifier over the NER label set.

To ablate the fine-tuning approach, we apply the feature-based approach by extracting the activations from one or more layers *without* fine-tuning any parameters of BERT. These contextual embeddings are used as input to a randomly initialized two-layer 768-dimensional BiLSTM before the classification layer.

Results are presented in Table 7. BERT_{LARGE} performs competitively with state-of-the-art methods. The best performing method concatenates the token representations from the top four hidden layers of the pre-trained Transformer, which is only 0.3 F1 behind fine-tuning the entire model. This demonstrates that BERT is effective for both fine-tuning and feature-based approaches.

6 Conclusion

Recent empirical improvements due to transfer learning with language models have demonstrated that rich, unsupervised pre-training is an integral part of many language understanding systems. In particular, these results enable even low-resource tasks to benefit from deep unidirectional architectures. Our major contribution is further generalizing these findings to deep *bidirectional* architectures, allowing the same pre-trained model to successfully tackle a broad set of NLP tasks.

关于增大预训练双向语言模型规模（从两层增至四层）对下游任务的影响，结果不一；Melamud 等人(2016)顺带提到，将隐藏维度从 200 增至 600 有所帮助，但进一步增至 1,000 并未带来更多提升。这两项先前研究都采用了基于特征的方法——我们推测，如果模型直接在下游任务上进行微调，且只使用极少量随机初始化的额外参数，那么即使下游任务数据非常有限，特定任务模型也能从规模更大、表达能力更强的预训练表示中受益。

5.3 基于 BERT 的特征方法

迄今为止介绍的所有 BERT 结果都采用了微调方法：在预训练模型上添加一个简单的分类层，并针对下游任务对所有参数进行联合微调。然而，基于特征的方法也有一定优势；这种方法从预训练模型中提取固定特征。首先，并非所有任务都能轻易地用 Transformer 编码器架构表示，因此需要添加特定于任务的模型架构。其次，预先计算一次训练数据的高成本表示，然后基于该表示使用成本较低的模型运行多次实验，可以显著节省计算资源。

在本节中，我们将 BERT 应用于 CoNLL-2003 命名实体识别（NER）任务，以比较这两种方法（Tjong Kim Sang and De Meulder, 2003）。在 BERT 的输入中，我们使用保留大小写的 WordPiece 模型，并纳入数据所提供的最大文档上下文。按照标准做法，我们将其表述为一个标注任务，但不使用 CRF

超参数				开发集准确率		
#L	#H	#A	语言模型（困惑度）	MNLI-m	MRPC	SST-2
3	768	12	5.84	77.9	79.8	88.4
6	768	3	5.24	80.6	82.2	90.7
6	768	12	4.68	81.9	84.8	91.3
12	768	12	3.99	84.4	86.7	92.9
12	1024	16	3.54	85.7	86.9	93.3
24	1024	16	3.23	86.6	87.8	93.7

表 6：对 BERT 模型规模的消融研究。#L = 层数；#H =隐藏层大小；#A = 注意力头数。“LM (ppl)”是留出训练数据的掩码语言模型困惑度。

系统	开发集 F1	测试集 F1
ELMo (Peters et al., 2018a)	95.7	92.2
CVT (Clark et al., 2018)	-	92.6
CSE (Akbik et al., 2018)	-	93.1
微调方法		
BERT _{LARGE}	96.6	92.8
BERT _{BASE}	96.4	92.4
基于特征的方法（BERT _{BASE} ）		
嵌入	91.0	-
倒数第二层隐藏状态	95.6	-
最后一层隐藏状态	94.9	-
最后四层隐藏状态的加权和	95.9	-
拼接最后四层隐藏层	96.1	-
对全部 12 层加权求和	95.5	-

表 7：CoNLL-2003 命名实体识别结果。超参数使用开发集进行选择。报告的开发集和测试集分数，是使用这些超参数进行 5 次随机重启后取平均得到的。

输出层。我们使用第一个子词元的表示作为输入，传递给针对 NER 标签集的词元级分类器。

为了消融微调方法，我们采用基于特征的方法，从 BERT 的一个或多个层中提取激活值而不 对 BERT 的任何参数进行微调。然后将这些上下文嵌入作为输入，送入分类层之前一个随机初始化的两层、每层 768 维的 BiLSTM。

结果见表7。BERT_{LARGE}的表现与最先进的方法相比毫不逊色。表现最佳的方法将预训练 Transformer 顶部四个隐藏层的词元表示拼接起来，其 F1 分数仅比对整个模型进行微调低 0.3。这表明 BERT 在微调和基于特征的方法中都很有效。

6 结论

近期，语言模型迁移学习带来的实证改进表明，丰富的无监督预训练是许多语言理解系统不可或缺的一部分。尤其是，这些成果使低资源任务也能从深度单向架构中受益。我们的主要贡献是进一步将这些发现推广到深度双向 架构，使同一个预训练模型能够成功应对广泛的 NLP 任务。

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Appendix for “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”

We organize the appendix into three sections:

- Additional implementation details for BERT are presented in Appendix A;

- Additional details for our experiments are presented in Appendix B; and

- Additional ablation studies are presented in Appendix C.

We present additional ablation studies for BERT including:

- Effect of Number of Training Steps; and
- Ablation for Different Masking Procedures.

A Additional Details for BERT

A.1 Illustration of the Pre-training Tasks

We provide examples of the pre-training tasks in the following.

Masked LM and the Masking Procedure Assuming the unlabeled sentence is `my dog is hairy`, and during the random masking procedure we chose the 4-th token (which corresponding to `hairy`), our masking procedure can be further illustrated by

- 80% of the time: Replace the word with the [MASK] token, e.g., `my dog is hairy` → `my dog is [MASK]`

- 10% of the time: Replace the word with a random word, e.g., `my dog is hairy` → `my dog is apple`

- 10% of the time: Keep the word unchanged, e.g., `my dog is hairy` → `my dog is hairy`. The purpose of this is to bias the representation towards the actual observed word.

The advantage of this procedure is that the Transformer encoder does not know which words it will be asked to predict or which have been replaced by random words, so it is forced to keep a distributional contextual representation of every input token. Additionally, because random replacement only occurs for 1.5% of all tokens (i.e., 10% of 15%), this does not seem to harm the model’s language understanding capability. In Section C.2, we evaluate the impact this procedure.

Compared to standard language model training, the masked LM only make predictions on 15% of tokens in each batch, which suggests that more pre-training steps may be required for the model

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Yukun Zhu, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. 2015. 对齐书籍与电影：通过观看电影和阅读书籍，实现类故事的视觉解释。载于IEEE国际计算机视觉会议论文集，第19–27 页。

《BERT：面向语言理解的深度双向Transformer 预训练》附录

我们将附录分为三个部分：

- BERT 的更多实现细节见附录A；

- 我们的实验的更多细节见附录 B；以及

- 更多消融研究见附录 C。

我们介绍了 BERT 的其他消融研究，包括：

- 训练步数的影响；以及– 不同掩码策略的消融实验。

A BERT 的其他细节

A.1 预训练任务示例

以下提供预训练任务的示例。

掩码语言模型（Masked LM）与掩码过程假设未标注句子是我的狗毛茸茸的，在随机掩码过程中，我们选择了第 4 个词元（对应毛茸茸），掩码过程可以进一步说明如下：

- 80有%的概率：将该词替换为[MASK]词元，例如：我的狗毛茸茸的 →我的狗 [MASK]

- 10有%的概率：将该词替换为一个随机词，例如：我的狗毛茸茸的 → 我的狗是苹果

- 10有%的概率：保持该词不变，例如，我的狗毛茸茸的 → 我的狗毛茸茸的。这样做的目的是使表示偏向于实际观察到的词。

这种做法的优点在于，Transformer 编码器并不知道会要求它预测哪些词，也不知道哪些词会被随机词替换，因此它不得不保留每个输入标记的分布式上下文表示。此外，由于随机替换只发生在所有标记的 1.5% 上（即 15% 的 10%），这似乎并未损害模型的语言理解能力。在第 C.2 节中，我们评估了这一做法的影响。

与标准语言模型训练相比，掩码语言模型在每个批次中仅对 15% 的词元进行预测，这意味着模型可能需要更多预训练步骤才能收敛

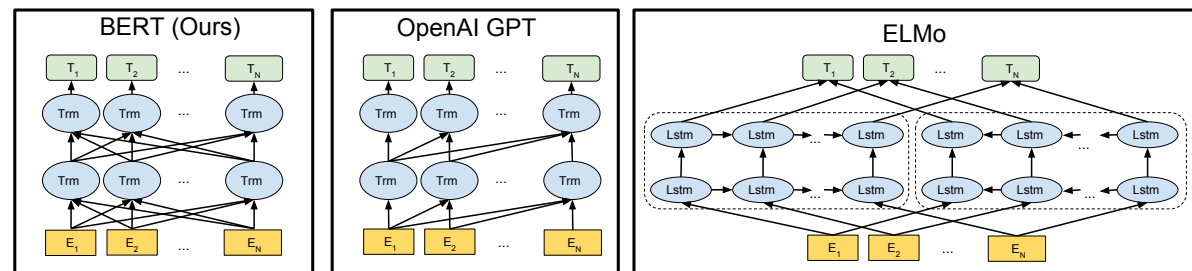


Figure 3: Differences in pre-training model architectures. BERT uses a bidirectional Transformer. OpenAI GPT uses a left-to-right Transformer. ELMo uses the concatenation of independently trained left-to-right and right-to-left LSTMs to generate features for downstream tasks. Among the three, only BERT representations are jointly conditioned on both left and right context in all layers. In addition to the architecture differences, BERT and OpenAI GPT are fine-tuning approaches, while ELMo is a feature-based approach.

to converge. In Section C.1 we demonstrate that MLM does converge marginally slower than a left-to-right model (which predicts every token), but the empirical improvements of the MLM model far outweigh the increased training cost.

Next Sentence Prediction The next sentence prediction task can be illustrated in the following examples.

Input = [CLS] the man went to [MASK] store [SEP]
he bought a gallon [MASK] milk [SEP]
Label = IsNext

Input = [CLS] the man [MASK] to the store [SEP]
penguin [MASK] are flight ##less birds [SEP]
Label = NotNext

A.2 Pre-training Procedure

To generate each training input sequence, we sample two spans of text from the corpus, which we refer to as “sentences” even though they are typically much longer than single sentences (but can be shorter also). The first sentence receives the A embedding and the second receives the B embedding. 50% of the time B is the actual next sentence that follows A and 50% of the time it is a random sentence, which is done for the “next sentence prediction” task. They are sampled such that the combined length is ≤ 512 tokens. The LM masking is applied after WordPiece tokenization with a uniform masking rate of 15%, and no special consideration given to partial word pieces.

We train with batch size of 256 sequences (256 sequences * 512 tokens = 128,000 tokens/batch) for 1,000,000 steps, which is approximately 40

epochs over the 3.3 billion word corpus. We use Adam with learning rate of $1e-4$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, L2 weight decay of 0.01, learning rate warmup over the first 10,000 steps, and linear decay of the learning rate. We use a dropout probability of 0.1 on all layers. We use a gelu activation (Hendrycks and Gimpel, 2016) rather than the standard relu, following OpenAI GPT. The training loss is the sum of the mean masked LM likelihood and the mean next sentence prediction likelihood.

Training of BERT_{BASE} was performed on 4 Cloud TPUs in Pod configuration (16 TPU chips total).¹³ Training of BERT_{LARGE} was performed on 16 Cloud TPUs (64 TPU chips total). Each pre-training took 4 days to complete.

Longer sequences are disproportionately expensive because attention is quadratic to the sequence length. To speed up pretraining in our experiments, we pre-train the model with sequence length of 128 for 90% of the steps. Then, we train the rest 10% of the steps of sequence of 512 to learn the positional embeddings.

A.3 Fine-tuning Procedure

For fine-tuning, most model hyperparameters are the same as in pre-training, with the exception of the batch size, learning rate, and number of training epochs. The dropout probability was always kept at 0.1. The optimal hyperparameter values are task-specific, but we found the following range of possible values to work well across all tasks:

- Batch size: 16, 32

¹³<https://cloudplatform.googleblog.com/2018/06/Cloud-TPU-now-offers-preemptible-pricing-and-global-availability.html>

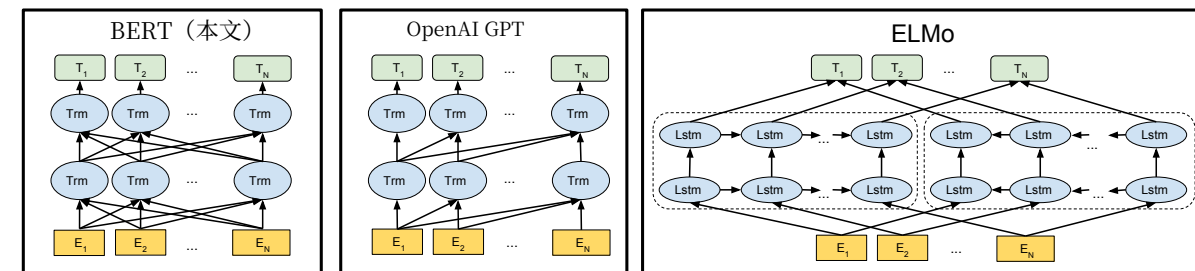


图 3: 预训练模型架构的差异。BERT 使用双向 Transformer。OpenAI GPT 使用从左到右的 Transformer。ELMo 将独立训练的从左到右和从右到左 LSTM 拼接起来，为下游任务生成特征。在这三种模型中，只有 BERT 的表示在所有层中都同时基于左侧和右侧上下文。除了架构上的差异，BERT 和 OpenAI GPT 采用微调方法，而 ELMo 采用基于特征的方法。

才能收敛。在第 C.1 节中，我们证明，MLM 的收敛速度确实略慢于从左到右的模型（后者会预测每个词元），但 MLM 模型带来的实证改进远远超过了增加的训练成本。

下一句预测 下一句预测任务可以通过以下示例来说明。

输入 = [CLS] 这个男人去了 [MASK] 商店 [SEP] 他买了一加仑
[MASK] 牛奶 [SEP] 标签 = IsNext 输入 = [CLS] 这
个男人 [MASK] 去商店 [SEP] 企鹅 [MASK] 是不会飞的鸟 [SEP]
标签 = NotNext

A.2 预训练流程

为了生成每个训练输入序列，我们从语料库中采样两个文本片段。尽管它们通常比单个句子长得多（但有时也会更短），我们仍将其称为“句子”。第一个句子使用 A 嵌入，第二个句子使用 B 嵌入。50% 的情况下，B 是紧接在 A 之后的实际下一句；另外 50% 的情况下，B 是一个随机句子，这样做是为了完成“下一句预测”任务。采样时会确保两者合并后的长度为 ≤ 512 个 token。LM 掩码在 WordPiece 分词之后应用，掩码率统一为 15%，且不对部分词片段进行特殊处理。

我们以 256 个序列为批次大小（256 个序列 * 512 个词元 = 128,000 个词元/批次）训练 1,000,000 步，约为 40

轮次，语料库包含 33 亿个词。我们使用 $\beta_1 = 0.9$, $\beta_2 = 0.999$, L2 权重衰减为 0.01, 学习率在前 10,000 步内预热，然后线性衰减。各层的 dropout 概率均为 0.1。我们使用 GELU 激活函数 (Hendrycks 和 Gimpel, 2016) 而非标准的 ReLU，遵循 OpenAI GPT。训练损失是掩码语言模型平均似然与下一句预测平均似然之和。

BERT_{BASE} 的训练使用了 4 个 Cloud TPU，采用 Pod 配置（共 16 个 TPU 芯片）。¹³ BERT_{LARGE} 的训练使用了 16 个 Cloud TPU（共 64 个 TPU 芯片）。每次预训练耗时 4 天。

较长的序列会带来不成比例的高昂计算开销，因为注意力机制的计算复杂度与序列长度的平方成正比。为了加快我们实验中的预训练速度，我们在 90% 的训练步骤中使用长度为 128 的序列预训练模型。然后，在剩余 10% 的训练步骤中使用长度为 512 的序列，以学习位置嵌入。

A.3 微调流程

对于微调，大多数模型超参数与预训练时相同，唯一区别是批量大小、学习率和训练轮数。dropout 概率始终保持为 0.1。最佳超参数值因任务而异，但我们发现，以下取值范围适用于所有任务：

- 批次大小: 16, 32

¹³<https://cloudplatform.googleblog.com/2018/06/Cloud-TPU-now-offers-preemptible-pricing-and-global-availability.html>

- **Learning rate (Adam):** 5e-5, 3e-5, 2e-5
- **Number of epochs:** 2, 3, 4

We also observed that large data sets (e.g., 100k+ labeled training examples) were far less sensitive to hyperparameter choice than small data sets. Fine-tuning is typically very fast, so it is reasonable to simply run an exhaustive search over the above parameters and choose the model that performs best on the development set.

A.4 Comparison of BERT, ELMo ,and OpenAI GPT

Here we studies the differences in recent popular representation learning models including ELMo, OpenAI GPT and BERT. The comparisons between the model architectures are shown visually in Figure 3. Note that in addition to the architecture differences, BERT and OpenAI GPT are fine-tuning approaches, while ELMo is a feature-based approach.

The most comparable existing pre-training method to BERT is OpenAI GPT, which trains a left-to-right Transformer LM on a large text corpus. In fact, many of the design decisions in BERT were intentionally made to make it as close to GPT as possible so that the two methods could be minimally compared. The core argument of this work is that the bi-directionality and the two pre-training tasks presented in Section 3.1 account for the majority of the empirical improvements, but we do note that there are several other differences between how BERT and GPT were trained:

- GPT is trained on the BooksCorpus (800M words); BERT is trained on the BooksCorpus (800M words) and Wikipedia (2,500M words).
- GPT uses a sentence separator ([SEP]) and classifier token ([CLS]) which are only introduced at fine-tuning time; BERT learns [SEP], [CLS] and sentence A/B embeddings during pre-training.
- GPT was trained for 1M steps with a batch size of 32,000 words; BERT was trained for 1M steps with a batch size of 128,000 words.
- GPT used the same learning rate of 5e-5 for all fine-tuning experiments; BERT chooses a task-specific fine-tuning learning rate which performs the best on the development set.

To isolate the effect of these differences, we perform ablation experiments in Section 5.1 which demonstrate that the majority of the improvements are in fact coming from the two pre-training tasks and the bidirectionality they enable.

A.5 Illustrations of Fine-tuning on Different Tasks

The illustration of fine-tuning BERT on different tasks can be seen in Figure 4. Our task-specific models are formed by incorporating BERT with one additional output layer, so a minimal number of parameters need to be learned from scratch. Among the tasks, (a) and (b) are sequence-level tasks while (c) and (d) are token-level tasks. In the figure, E represents the input embedding, T_i represents the contextual representation of token i , [CLS] is the special symbol for classification output, and [SEP] is the special symbol to separate non-consecutive token sequences.

B Detailed Experimental Setup

B.1 Detailed Descriptions for the GLUE Benchmark Experiments.

Our GLUE results in Table1 are obtained from <https://gluebenchmark.com/leaderboard> and <https://blog.openai.com/language-unsupervised>. The GLUE benchmark includes the following datasets, the descriptions of which were originally summarized in Wang et al. (2018a):

MNLI Multi-Genre Natural Language Inference is a large-scale, crowdsourced entailment classification task (Williams et al., 2018). Given a pair of sentences, the goal is to predict whether the second sentence is an *entailment*, *contradiction*, or *neutral* with respect to the first one.

QQP Quora Question Pairs is a binary classification task where the goal is to determine if two questions asked on Quora are semantically equivalent (Chen et al., 2018).

QNLI Question Natural Language Inference is a version of the Stanford Question Answering Dataset (Rajpurkar et al., 2016) which has been converted to a binary classification task (Wang et al., 2018a). The positive examples are (question, sentence) pairs which do contain the correct answer, and the negative examples are (question, sentence) from the same paragraph which do not contain the answer.

- **学习率 (Adam) :** 5e-5, 3e-5, 2e-5
- **训练轮数:** 2, 3, 4

我们还发现, 大型数据集 (例如, 100k + 个带标签的训练样本) 对超参数选择的敏感度远低于小型数据集。微调通常非常快, 因此直接对上述参数进行穷举搜索, 并选择在开发集上表现最佳的模型, 是合理的做法。

A.4 BERT、ELMo 和 OpenAI GPT 的比较

本文研究了近期流行的表示学习模型之间的差异, 包括 ELMo、OpenAI GPT 和 BERT。图3以直观的方式展示了模型架构之间的差异。请注意, 除了架构差异之外, BERT 和 OpenAI GPT 采用微调方法, 而 ELMo 则采用基于特征的方法。

与 BERT 最具可比性的现有预训练方法是 OpenAI GPT, 它在大型文本语料库上训练从左到右的 Transformer 语言模型。事实上, BERT 的许多设计决策都是有意为之, 目的是使其尽可能接近 GPT, 从而能够尽量公平地比较这两种方法。本文的核心论点是, 双向性以及第3.1 节介绍的两个预训练任务解释了大部分实证效果的提升, 但我们也注意到, BERT 和 GPT 的训练方式还有其他一些差异:

- GPT 在 BooksCorpus (8 亿个词) 上进行训练; BERT 在 BooksCorpus (8 亿个词) 和维基百科 (25 亿个词) 上进行训练。
- GPT 使用句子分隔符 ([SEP]) 和分类器标记 ([CLS]), 这些标记仅在微调时引入; BERT 则在预训练期间学习[SEP], [CLS] 以及句子 A/B 嵌入。
- GPT 训练了 100 万步, 批次大小为 32,000 个词; BERT 训练了 100 万步, 批次大小为 128,000 个词。
- GPT 在所有微调实验中都使用相同的学习率 5e-5; BERT 则针对不同任务选择在开发集上效果最佳的微调学习率。

为单独考察这些差异的影响, 我们在第 5.1 节进行了消融实验, 结果表明, 大部分改进实际上都来自这两个预训练任务及其所实现的双向性。

A.5 不同任务上的微调示例

BERT 在不同任务上进行微调的示例如图 4所示。我们的特定任务模型通过将 BERT 与一个额外的输出层结合而构建, 因此只需从头学习少量参数。其中, (a) 和 (b) 是序列级任务, 而 (c) 和 (d) 是词元级任务。在图中, E 表示输入嵌入, T_i 表示词元 i , [CLS] 是用于分类输出的特殊符号, 而 [SEP] 是用于分隔非连续词元序列的特殊符号。

B 详细实验设置

B.1 GLUE 基准实验的详细说明。

我们的 GLUE 结果见表1 , 数据来自 <https://gluebenchmark.com/leaderboard>和 <https://blog.openai.com/language-unsupervised>。GLUE 基准包含以下数据集, 其描述最初由Wang et al.(2018a) 总结:

MNLI 多体裁自然语言推理是一项大规模、通过众包完成的蕴含分类任务 (Williams et al., 2018)。给定一对句子, 目标是预测第二个句子相对于第一个句子是蕴含、矛盾还是中立。

QQP Quora 问题对是一项二元分类任务, 目标是判断在 Quora 上提出的两个问题在语义上是否等价 (Chen et al., 2018)。

QNLI 问题自然语言推理是斯坦福问答数据集 (Rajpurkaret al., 2016) 的一个版本, 已被转换为二元分类任务 (Wanget al.,2018a)。正例是包含正确答案的 (问题, 句子) 对, 负例则是来自同一段落、但不包含答案的 (问题, 句子) 对。

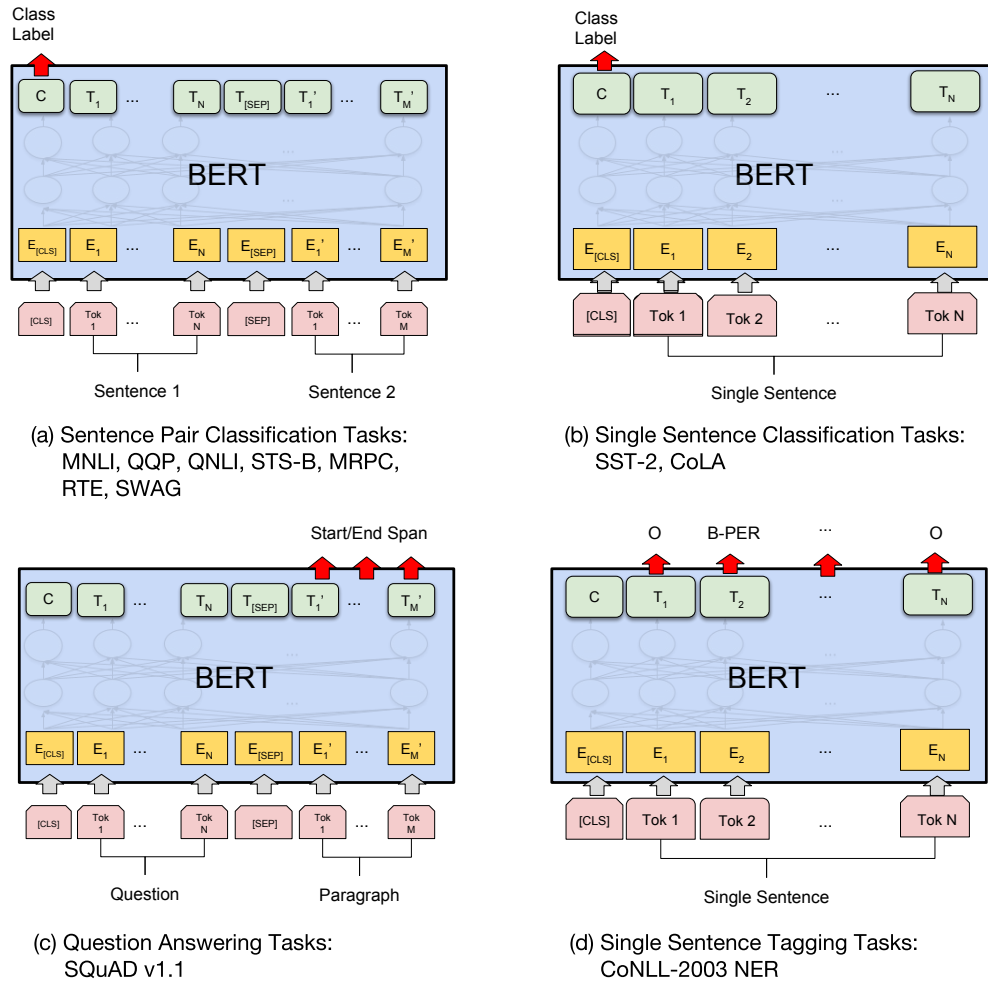


Figure 4: Illustrations of Fine-tuning BERT on Different Tasks.

SST-2 The Stanford Sentiment Treebank is a binary single-sentence classification task consisting of sentences extracted from movie reviews with human annotations of their sentiment (Socher et al., 2013).

CoLA The Corpus of Linguistic Acceptability is a binary single-sentence classification task, where the goal is to predict whether an English sentence is linguistically “acceptable” or not (Warstadt et al., 2018).

STS-B The Semantic Textual Similarity Benchmark is a collection of sentence pairs drawn from news headlines and other sources (Cer et al., 2017). They were annotated with a score from 1 to 5 denoting how similar the two sentences are in terms of semantic meaning.

MRPC Microsoft Research Paraphrase Corpus consists of sentence pairs automatically extracted from online news sources, with human annotations

for whether the sentences in the pair are semantically equivalent (Dolan and Brockett, 2005).

RTE Recognizing Textual Entailment is a binary entailment task similar to MNLI, but with much less training data (Bentivogli et al., 2009).¹⁴

WNLI Winograd NLI is a small natural language inference dataset (Levesque et al., 2011). The GLUE webpage notes that there are issues with the construction of this dataset,¹⁵ and every trained system that’s been submitted to GLUE has performed worse than the 65.1 baseline accuracy of predicting the majority class. We therefore exclude this set to be fair to OpenAI GPT. For our GLUE submission, we always predicted the ma-

¹⁴Note that we only report single-task fine-tuning results in this paper. A multitask fine-tuning approach could potentially push the performance even further. For example, we did observe substantial improvements on RTE from multitask training with MNLI.

¹⁵<https://gluebenchmark.com/faq>

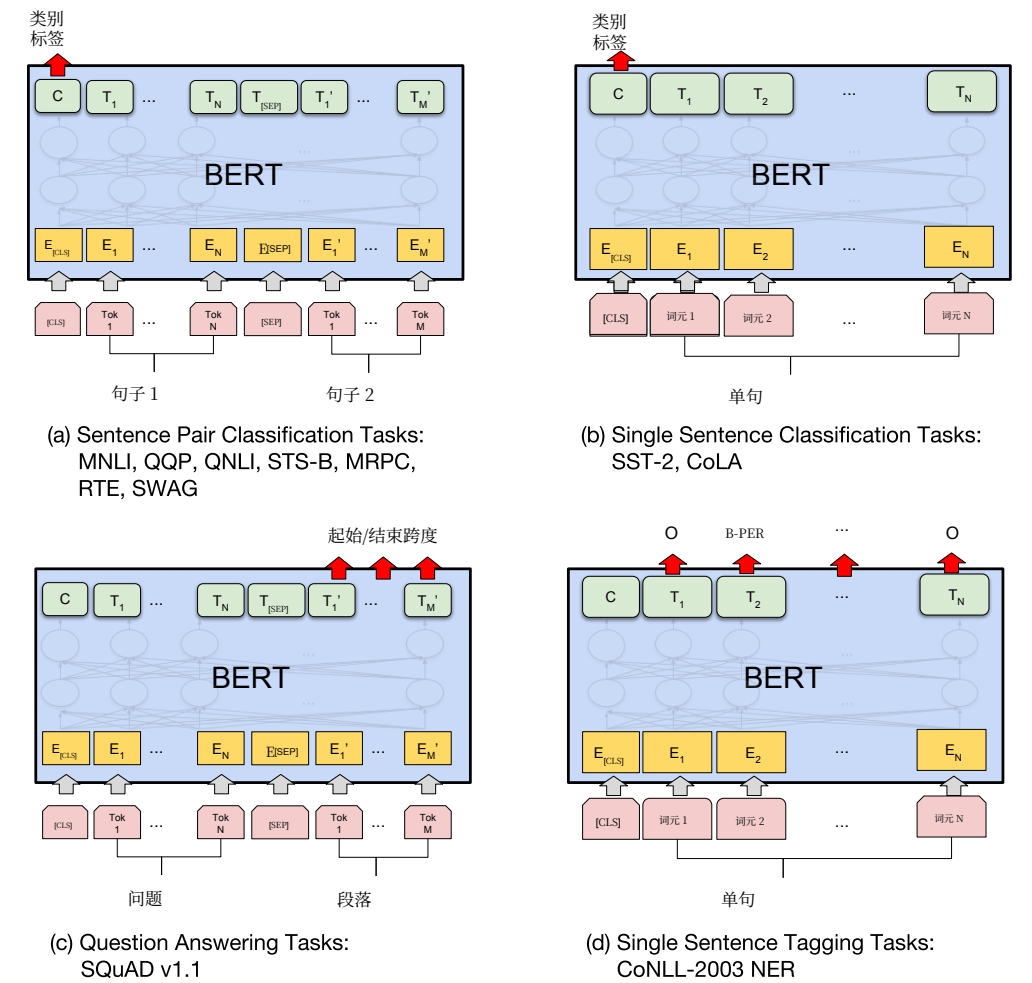


图 4：在不同任务上微调 BERT 的示意图。

SST-2 斯坦福情感树库是一项二元单句分类任务，其中的句子摘自电影评论，并由人工标注了情感倾向 (Socher et al., 2013)。

CoLA 语言可接受性语料库是一种二元单句分类任务，其目标是预测一个英语句子在语言学上是否“可接受” (Warstadt et al., 2018)。

STS-B 语义文本相似度基准由从新闻标题和其他来源收集的句子对组成 (Cer et al., 2017)。标注者为每对句子赋予 1 到 5 分，以表示这两个句子在语义上的相似程度。

MRPC 微软研究院释义语料库由从在线新闻来源自动提取的句子对组成，并由人工标注

用于判断句子对中的句子在语义上是否等价 (Dolan and Brockett, 2005)。

RTE 文本蕴含识别是一项二元蕴含任务，与 MNLI 类似，但训练数据少得多 (Bentivogli et al., 2009)。¹⁴

WNLI Winograd NLI 是一个小型自然语言推理数据集 (Levesque et al., 2011)。GLUE 网页指出，该数据集的构建存在一些问题，¹⁵ 而且，所有提交至 GLUE 的训练系统，其表现都不如始终预测多数类所得的 65.1 基准准确率。因此，为公平对待 OpenAI GPT，我们排除了这个数据集。在我们的 GLUE 提交中，我们始终预测多

¹⁴请注意，本文仅报告单任务微调的结果。多任务微调方法或许还能进一步提升性能。例如，我们确实观察到，通过与 MNLI 进行多任务训练，RTE 的表现有了显著提升。¹⁵<https://gluebenchmark.com/faq>

jority class.

C Additional Ablation Studies

C.1 Effect of Number of Training Steps

Figure 5 presents MNLI Dev accuracy after fine-tuning from a checkpoint that has been pre-trained for k steps. This allows us to answer the following questions:

1. Question: Does BERT really need such a large amount of pre-training (128,000 words/batch * 1,000,000 steps) to achieve high fine-tuning accuracy?
Answer: Yes, BERT_{BASE} achieves almost 1.0% additional accuracy on MNLI when trained on 1M steps compared to 500k steps.
2. Question: Does MLM pre-training converge slower than LTR pre-training, since only 15% of words are predicted in each batch rather than every word?

Answer: The MLM model does converge slightly slower than the LTR model. However, in terms of absolute accuracy the MLM model begins to outperform the LTR model almost immediately.

C.2 Ablation for Different Masking Procedures

In Section 3.1, we mention that BERT uses a mixed strategy for masking the target tokens when pre-training with the masked language model (MLM) objective. The following is an ablation study to evaluate the effect of different masking strategies.

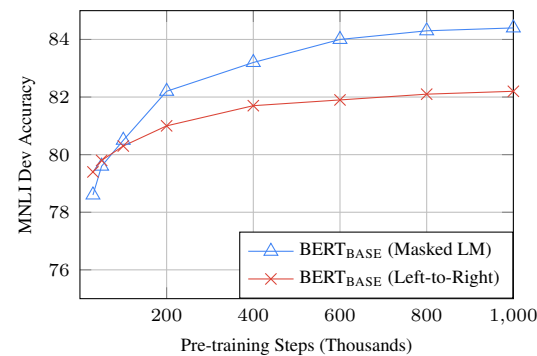


Figure 5: Ablation over number of training steps. This shows the MNLI accuracy after fine-tuning, starting from model parameters that have been pre-trained for k steps. The x-axis is the value of k .

Note that the purpose of the masking strategies is to reduce the mismatch between pre-training and fine-tuning, as the [MASK] symbol never appears during the fine-tuning stage. We report the Dev results for both MNLI and NER. For NER, we report both fine-tuning and feature-based approaches, as we expect the mismatch will be amplified for the feature-based approach as the model will not have the chance to adjust the representations.

Masking Rates			Dev Set Results		
MASK	SAME	RND	MNLI Fine-tune	NER	
				Fine-tune	Feature-based
80%	10%	10%	84.2	95.4	94.9
100%	0%	0%	84.3	94.9	94.0
80%	0%	20%	84.1	95.2	94.6
80%	20%	0%	84.4	95.2	94.7
0%	20%	80%	83.7	94.8	94.6
0%	0%	100%	83.6	94.9	94.6

Table 8: Ablation over different masking strategies.

The results are presented in Table 8. In the table, MASK means that we replace the target token with the [MASK] symbol for MLM; SAME means that we keep the target token as is; RND means that we replace the target token with another random token.

The numbers in the left part of the table represent the probabilities of the specific strategies used during MLM pre-training (BERT uses 80%, 10%, 10%). The right part of the paper represents the Dev set results. For the feature-based approach, we concatenate the last 4 layers of BERT as the features, which was shown to be the best approach in Section 5.3.

From the table it can be seen that fine-tuning is surprisingly robust to different masking strategies. However, as expected, using only the MASK strategy was problematic when applying the feature-based approach to NER. Interestingly, using only the RND strategy performs much worse than our strategy as well.

多数类。

C 附加消融研究

C.1 训练步数的影响

图5展示了从预训练步数为 k 步的检查点微调后的 MNLI 开发集准确率。这使我们能够回答以下问题：

1. 问题：BERT 是否真的需要如此大量的预训练（128,000 词/批次 * 1,000,000 步），才能获得较高的微调准确率？
回答：训练 1M 步而非 500k 步时，BERT_{BASE} 在 MNLI 上的准确率几乎额外提升了 1.0%。
2. 问题：由于每个批次中只预测 15% 的词，而不是预测所有词，MLM 预训练是否比 LTR 预训练收敛得更慢？

答案：MLM 模型的收敛速度确实比 LTR 模型稍慢。不过，从绝对准确率来看，MLM 模型几乎从一开始就优于 LTR 模型。

C.2 不同掩码策略的消融实验

在第 3.1 节中，我们提到，BERT 在使用掩码语言模型（MLM）目标进行预训练时，会采用混合策略对目标词元进行掩码。以下消融研究旨在评估不同掩码策略的效果。

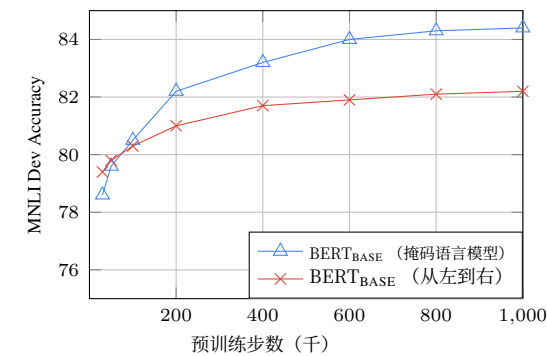


图 5：不同训练步数的消融实验。图中展示了经过微调后的 MNLI 准确率，模型参数来自已预训练 k 步的模型。横轴为 k 的取值。

请注意，采用这些掩码策略的目的是减少预训练和微调之间的不匹配，因为在微调阶段不会出现 [MASK] 符号。我们报告了 MNLI 和 NER 的开发集结果。对于 NER，我们报告了微调 and 基于特征的方法的结果，因为我们预计，对于基于特征的方法，这种不匹配会更加明显：模型没有机会调整其表示。

掩码比例			开发集结果		
掩码相同		RND	MNLI 微调	NER 基于特征	
80%	10%	10%	84.2	95.4	94.9
100%	0%	0%	84.3	94.9	94.0
80%	0%	20%	84.1	95.2	94.6
80%	20%	0%	84.4	95.2	94.7
0%	20%	80%	83.7	94.8	94.6
0%	0%	100%	83.6	94.9	94.6

表 8：不同掩码策略的消融实验。

结果见表8。在表中，MASK 表示在 MLM 中用 [MASK] 符号替换目标词元；SAME 表示保留目标词元不变；RND 表示用另一个随机词元替换目标词元。

表格左侧的数字表示 MLM 预训练期间采用特定策略的概率（BERT 使用 80%、10%、10%）。表格右侧列出开发集的结果。对于基于特征的方法，我们将 BERT 的最后 4 层拼接作为特征，这已在第 5.3 节中证明是最佳方法。

从表中可以看出，微调对不同的掩码策略有着出乎意料的鲁棒性。不过，正如预期的那样，在将基于特征的方法应用于 NER 时，仅使用 MASK 策略会带来问题。有趣的是，仅使用 RND 策略的效果也远不如我们的策略。